Renewable and Sustainable Energy Reviews 226 (2026) 116294

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: www.elsevier.com/locate/rser

Mathematical modeling of heat and mass transfer in metal hydride hydrogen storage systems: A comprehensive review

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ARTICLE INFO

Keywords:
Hydrogen storage
Metal hydrides
Mathematical modeling
Modeling review
Reactive porous media
Heat and mass transfer

ABSTRACT

Metal hydrides (MHs) are among the most promising materials for safe, compact, and reversible hydrogen storage, but their deployment is constrained by slow kinetics and thermal management challenges. Since MH performance is strongly governed by coupled heat and mass transfer processes, mathematical modeling has become essential for optimizing and designing storage systems. This review addresses a critical gap since the last comprehensive review in 2016 by synthesizing state-of-the-art mathematical modeling approaches for heat, mass, and momentum transfer in MH reactors. Starting from effective medium theory, we formulate macroscopic conservation equations and critically compare local thermal equilibrium (LTE) and non-equilibrium (LTNE) models. LTE models are computationally efficient but may underpredict wall heat fluxes, while LTNE models enhance accuracy at higher computational cost. We analyze empirical equilibrium pressure relations, reaction kinetics, reactor geometries, boundary conditions, and thermal management strategies, including phase change materials (PCMs) and heat transfer fluids (HTF). While metal foam integration can enhance charging rates by up to 65 %, phase change materials (PCMs) can reduce hydrogen absorption time by 60.2 % in metal hydride reactors. By consolidating theoretical and numerical perspectives, and comparing the trade-offs between various modeling approaches, this review identifies limitations and outlines future research directions to accelerate the design and deployment of efficient solid-state hydrogen storage technologies.

NOMENCLATURE

Symbol	Definition
A	Parameter in P-C-T equation, area [m^2]/Hydride forming element
a_s	specific surface area
B	Parameter in P-C-T equation [K]/Non hydride forming element
C_p	Specific heat capacity [J/(kg K)]
E	Activation energy [J/mol]
ε	Porosity
f	Cooling fluid
g	Gaseous phase (hydrogen)
H	Reaction enthalpy [J/mol H_2], reaction heat of formation [J/kg]
h	Convective heat transfer coefficient [W/(m^2 K)]
H/M	Hydrogen to metal atomic ratio
k	Reaction rate constant [s^{-1}]
K	Permeability [m^2]
liq	Liquid
λ	Thermal conductivity [W/(m K)]
m	Hydrogen mass absorbed [kg]
$\dot{m}$	Mass source term of reaction [kg/(m^3 s)]

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Symbol	Definition
M	Molecular weight [kg mol^{-1}]/Metal
MH	Metal hydride
N	Number of moles of hydrogen gas
P	Pressure [Pa]
P-C-T/PCT	Pressure-composition-temperature relationship
PCM	Phase change material
q	Mass flow rate [kg/s]
R	Universal gas constant [J/(mol K)]
R_g	General gas constant [J/(mol K)]
r	Radial coordinate [m]
S	Source term
s	Solid phase
sol	Solution
T	Temperature [K]
t	Time [s]
$U/\vec{U}/u$	Gas velocity [m/s]/Darcy's velocity
W/w	Mass [kg]

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https://doi.org/10.1016/j.rser.2025.116294
Received 9 July 2025; Received in revised form 23 August 2025; Accepted 11 September 2025
Available online 19 September 2025
1364-0321/

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Symbol	Definition
X	Reacted fraction of absorbed hydrogen
z	Axial coordinate [m]
α	Hydrogen-free metal phase
β	Hysteresis factor in P-C-T equation/Saturated hydride phase
μ	Dynamic viscosity [Pa.s]
τ	Stress tensor
τ_p	Relative expansion factor
ϕ	Plateau flatness factor in P-C-T equation
ϕ_0	Plateau flatness factor in P-C-T equation at reference state

1. Introduction

Growing concerns over the detrimental impacts of greenhouse gas emissions on climate change and global warming have prompted a significant push towards discovering sustainable and environmentally friendly energy alternatives. Hydrogen, due to its natural abundance and impressive energy density per unit mass, has emerged as a transformative candidate to replace conventional fuels [1]. However, the practical utilization of hydrogen faces a critical challenge in the form of storage, primarily due to its low energy density per unit volume, necessitating substantial space under ambient conditions. The most mature and conventional method for hydrogen storage is to compress it at high pressures, typically ranging from 500 to 700 bars. However, storing hydrogen in its gaseous state results in low storage density as shown in Table 1 [2]. Additionally, the use of high pressure introduces safety concerns, given that hydrogen is highly flammable. The second approach involving liquification of hydrogen under cryogenic conditions (−253 K at 1 atm) is subjected to additional costs associated with pressure containment and cryogenic temperature maintenance [3,4]. The volumetric and gravimetric densities of hydrogen in different forms under varying pressure and temperature conditions are shown in Table 1 [2,5].

In this context, metal hydrides (MH) have gained attention in the research community for their unique ability to store considerable amounts of hydrogen in solid phase, offering both safety and long-term stability [2,6,7]. Extensively researched since the 1970s, metal hydrides have demonstrated potential for hydrogen storage, thereby addressing some of the limitations of gaseous or liquid storage methods [7]. While MH systems are often considered safer, few studies provide system-level quantitative risk metrics, which require probabilistic and consequence analyses [8,9]. Key metrics include annualized rupture or large-release frequency, conditional ignition probability with thermal or overpressure consequences, and life-cycle risk indicators. Methodological guidance and hydride-specific event/fault-tree models are available for implementing such analyses [8–11].

The effective design of metal hydride storage systems requires rigorous mathematical modeling that can serve as a powerful tool for predicting the relationships between various influential factors, ultimately optimizing the performance of MH storage systems. Numerical models for metal hydride hydrogen storage systems have been developed and reviewed in existing literature [12,13]. These computational models are utilized to analyze one-dimensional [14–17], two-dimensional [5,18–23], and three-dimensional [24–28] systems. These models traditionally incorporate heat and mass balance equations alongside empirically derived reaction kinetics, and serve to analyze the thermal behavior, equilibrium conditions, and reaction kinetics of MH reactors. Notably, the three-dimensional model, while providing more detailed insights, demands significantly more time and computing capacity. Consequently, it is frequently utilized to explore phenomena that two-dimensional models may inadequately represent or overlook [5,27].

In 1995, Jemni and Nasrallah [23] conducted extensive experimental investigations to analyze the critical parameters influencing hydrogen absorption in an MH reactor. Their findings questioned the validity of the local thermal equilibrium assumption between the solid-phase hydride and the gaseous-phase hydrogen. In a later work published in 1997, Nasrallah and Jemni [29] reported contradictory results, concluding that disregarding the temperature gradients between the two phases had a minimal effect on the reacted fraction profile of the MH reactor during hydrogen sorption process. Following their work, almost all the subsequent studies adopted the assumption of local thermal equilibrium to simplify their models and reduce computational demands [7,30–32]. Several studies neglected the impact of advection-driven heat transport, and pressure gradients within the bed, citing the low hydrogen flow velocity within porous beds as justification [7,29,33,34]. Furthermore, some models assumed that the equilibrium pressure depended solely on temperature, overlooking its relationship with the hydrogen-to-metal atomic ratio (H/M) [13,24,35–38].

Mazzucco and Rokni [39] developed a 1D model to predict the absorbed hydrogen weight fraction and MH bed temperature, featuring a U-shaped aluminum coolant tube with axial symmetry embedded in the MH bed. Darzi et al. [40] employed a 2D model to study the heating and cooling of a cylindrical $LaNi_5$ MH tank, incorporating an annular jacket filled with phase-change material. Minko et al. [41] introduced a pore-scale model to visualize heat and mass transfer in a MH reactor with aluminum foam. Sekhar et al. [42] presented a 3D numerical model to evaluate hydrogen uptake performance with different heat exchanger configurations. Kyoung et al. [43] developed a 3D computational fluid dynamics (CFD) model to investigate reaction kinetics and heat and mass transfer during hydrogen desorption. Mohammadshahi et al. [44,45] developed a three-dimensional computational model for absorption-desorption cycle based on the Hardy and Anton mathematical framework [27] and examined the impact of various parameters on MH tank performance. Similarly, Mohammadshahi et al. [46] validated their model with local hydrogen concentration measurements obtained through volume-resolved neutron powder diffraction. Most published models rely on finite-element analysis and are predominantly two-dimensional.

The absorption and desorption of hydrogen in the porous MH bed are, respectively, exothermic and endothermic processes. Approximately 10–20 % of the high heating value (HHV) of hydrogen is released or absorbed during these reversible sorption processes, depending on the nature of the hydride bed, as well as the pressure, and temperature profiles [47,48]. Effective thermal management is crucial to maintain the operation temperature within the desired range, especially for removing the excess heat generated during the exothermic absorption process [31]. This not only sustains the absorption rate (charging rate) but also optimizes the utilization of MH storage capacity and overall system efficiency. Conversely, the endothermic desorption (discharging) process requires a continuous supply of thermal energy to the bed to maintain the hydrogen discharge rate and meet the energy demand from the fuel cell [31,47]. Given the significant impact of heat on sorption reaction rates, thermal management techniques play a vital role in the

Table 1
Comparison of hydrogen storage densities in different states under varying conditions [2,5].

Hydrogen storage forms and conditions	Volumetric hydrogen density (Theoretical limit) [kg H_2/m^3 system]	Gravimetric hydrogen density [kg H_2/kg system]
Compressed hydrogen at 25 °C and 50 MPa	27 (30.8)	4
Liquefied hydrogen at −253 °C and 0.1 MPa	40 (71)	3
Metal hydride at 25 °C and 0.1 MPa	50 (110)	1.2
Complex hydrides at 150 °C and 0.1 MPa	50 (120)	4
Metals at 25 °C and 0.1 MPa (Zn (H_2O))	90	3

design of MH-based hydrogen storage systems [49,50]. Despite the abundance of different thermal models, a review paper covering the mathematical modeling of these models for metal hydride systems is notably absent in the existing literature.

It is important to consolidate diverse approaches for heat, mass, and momentum transport to comprehend the advancements and latest trends in modeling MH hydrogen storage systems. Several existing review papers have concentrated on MH reactor and thermal management systems, with a primary focus on experimental and technical aspects related to the tank designs and cooling system configurations, MH materials, heat transfer challenges and enhancements [1,3,30,51–55].

Despite advancements in metal hydride (MH) hydrogen storage systems, key challenges persist, including the computational complexity of three-dimensional models, oversimplifications in local thermal equilibrium assumptions, and the lack of comprehensive comparative analyses of modeling approaches. This review addresses the critical gap in the literature since the last comprehensive review in 2016 [7] by providing an in-depth synthesis of mathematical modeling techniques for MH reactors. Its scope encompasses heat, mass, and momentum transfer models, equilibrium pressure relations, reaction kinetics, and thermal management strategies, with a focus on comparing local thermal equilibrium and non-equilibrium frameworks. The objectives are to consolidate diverse modeling approaches, critically evaluate their assumptions and limitations, and outline future research directions to enhance model accuracy and support the design of efficient, scalable MH-based hydrogen storage systems.

Furthermore, the paper comprehensively discusses essential geometrical and operational parameters that have been considered in prior research concerning MH tanks and thermal management systems. It evaluates equilibrium pressure, reaction kinetics, thermal management strategies, and various reactor configurations and designs. In short, this review fills a noticeable void in the literature, providing an extensive and comparative analysis of different theoretical aspects related to mathematical modeling of MH hydrogen storage systems.

2. Mathematical model

2.1. Effective medium theory: from microscopic to macroscopic scale

A metal hydride bed is composed of a heterogeneous two-phase system, consisting of solid MH particles and hydrogen in the gaseous phase. The governing equations for energy, mass, and momentum transfer in this reactive porous medium are formulated by scaling up from the microscopic level to the macroscopic level using the framework of effective medium theory [5,56].

At the microscopic scale, the system's details are considered at a very fine level, where the averaging volume (ω) is significantly smaller than the volumes of individual pores. This microscopic perspective captures the detailed interactions within the pores. However, for practical modeling and computational purposes, we shift to a macroscopic scale. At this macroscopic scale, the volumes of individual pores become negligible compared to the averaging volume. This transition effectively allows for the representation of inherently discontinuous medium (with distinct solid and gas phases) as a hypothetical continuous medium [34,56]. This conceptual shift from a microscopic to a macroscopic perspective is achieved by averaging the properties and behaviors over a representative elementary volume (REV). The REV is large enough to encompass multiple pores and particles, thereby smoothing out the discontinuities.

Each term in the macroscopic-scale continuum equations is formulated by averaging its corresponding term from the microscopic equations over this REV. The averaging volume for a porous material with solid and gaseous phases is demonstrated by Fig. 1. For a specific microscopic function ($\overline{\phi}$), the average is defined by Eq. (1). Similarly, Eq. (2) shows its intrinsic average within phase i in the multi-phase porous medium, with ω_i representing the individual volume of phase i in averaging volume ω.

$$\overline{\phi} = 1 \Big/ \omega \int_{\omega} \phi \, d\omega \tag{1}$$

$$\overline{\phi_i} = 1 \Big/ \omega_i \int_{\omega_i} \phi_i \, d\omega \tag{2}$$

2.2. Macroscopic differential equations

2.2.1. Model assumptions and metal hydride basics

The macroscopic PDE model for the hypothetical continuous medium is derived by applying the averaging process to the corresponding microscopic equations and incorporating closure assumptions across the averaging volume ω. The microscopic equations are based on the mass, energy, and momentum balance equations for each phase. The mathematical framework for describing the multiphysics sorption processes in a metal hydride (MH) reactor can be developed by incorporating the following simplifying assumptions:

i. Given that the operating conditions are near ambient conditions for most hydrides, the hydrogen gas within the metal hydride (MH) bed can be approximated as an ideal gas, enabling the use of the ideal gas law to establish a relationship between density and pressure.
ii. Radiative heat transfer is disregarded in the system due to its operation within a moderate temperature range, where the local temperature differences between neighboring MH particles are minimal relative to the absolute temperature, making the contribution of radiation to overall heat transfer negligible [5,57].
iii. The MH bed is assumed to be isotropic with homogeneous porosity [58].
iv. The volumetric expansion/compression, viscous dissipation, and variations in porosity of the bed during absorption/desorption are considered negligible [59].

The medium's thermal behavior can be modeled either under local

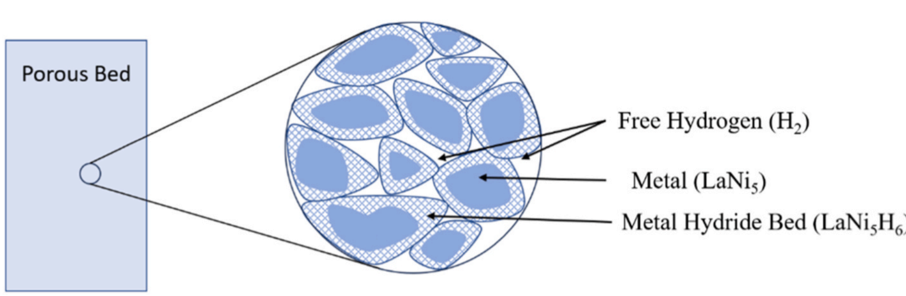

Fig. 1. Averaging volume for porous MH bed with solid and gas phases.

thermal equilibrium (LTE), where both phases are assumed at the same temperature [5,60], or under local thermal non-equilibrium (LTNE), which requires distinct temperature equations for each phase [23,29]. This distinction is crucial when formulating the energy conservation model, as it directly impacts the treatment of heat generation and transfer during hydrogen-metal interactions. The detailed heat transfer equations will be provided in the energy conservation section of the paper.

The reversible reaction in Eq. (3) describes the interaction between hydrogen and metal atoms (M) that occurs during absorption and desorption processes. These reactions are inherently exothermic or endothermic, with the reaction enthalpy, ΔH, varying based on the specific materials involved [61,62].

$$M(s) + \frac{n}{2}H_2(g) \rightleftharpoons MH_n(s) + \Delta H \tag{3}$$

where n represents the hydrogen-to-metal atomic ratio (H/M). On a mass basis, the total mass of hydrogen absorbed during this reaction can be written as:

$$w = \frac{nM_H}{M_M + nM_H} \tag{4}$$

where M_H is the atomic mass of hydrogen, and M_M represents the average atomic mass of the metal atoms. If the maximum hydrogen uptake (at saturation) is expressed as $\left(\frac{H}{M}\right)_{max}$ in atomic terms or w_{max} in mass terms, the reacted fraction X can be given by Eq. (1e). The reacted fraction (X) is the parameter [60,63,64] that quantifies the H/M on a scale of 0 (for hydrogen-free metal bed) to 1 (for saturated MH bed). Some studies refer to the hydrogen-free metal as the dilute or α-phase and the saturated metal hydride bed as the concentrated or β-phase [57,65]. Further details on reacted fraction will be provided in the reaction kinetics section of the mathematical model.

$$X = \frac{w}{w_{max}} = \frac{\left(\frac{H}{M}\right)}{\left(\frac{H}{M}\right)_{max}} \tag{5}$$

Different materials have been studied extensively for their hydrogen storage capabilities, leading to the discovery of various intermetallic hydrides with improved thermodynamic and kinetic properties [66]. Around the 1970s, hydrides with significantly lower hydrogen reaction enthalpies, such as $LaNi_5$, FeTi, and Mg_2Ni, were identified [67]. For instance, while pure high-temperature hydride LaH_2 requires approximately 1300 °C to achieve a desorption pressure of 2 bar, $LaNi_5H_6$ reaches the same plateau pressure at just 20 °C, with a reaction enthalpy of $\Delta H = 30.9\ kJ\ mol^{-1}\ H_2$ [1,67]. Since then, several hundred intermetallic hydrides have been reported, with key compositional types outlined in Table 2. Scarpati et al. [68] provided a detailed comparative analysis of the PCT plots of many materials corresponding to the intermetallic compositions listed in Table 2. These hydrides generally consist of a high-temperature hydride-forming element (A) combined with a non-hydride-forming element (B). Fig. 2 illustrates the periodic table, highlighting hydride-forming and non-hydride-forming elements alongside their respective enthalpy changes.

Table 2
Classification of metal hydrides based on their composition and associated compounds [67].

Compositional Category	A	B	Metal Hydrides
AB	Ti, Zr	Ni, Fe	TiFe, TiNi, ZrNi
A_2B	Mg, Zr	Ni, Fe, Co	Mg_2Co, Mg_2Ni, Zr_2Fe
AB_2	La, Ti, Y, Zr	V, Cr, Mn, Fe, Ni	$LaNi_2$, $TiMn_2$, YMn_2, YNi_2, $ZrCr_2$, $ZrMn_2$, ZrV_2
AB_3	La, Mg, Y	Ni, Co	$LaCo_3$, $LaMg_2Ni_9$, YNi_3
AB_5	Ca, La, Rare Earth	Ni, Cu, Co, Pt, Fe	$CaNi_5$, $CeNi_5$, $LaCu_5$, $LaFe_5$ $LaNi_5$, $LaPt_5$

Extensive research has been conducted to investigate the properties, mechanisms, and applications of metal hydrides, resulting in numerous review articles. These reviews provide a wealth of knowledge on the advancements in material selection, thermodynamic properties, and hydrogen sorption kinetics. Readers are referred to the following comprehensive review articles for further insights [1,66,68–72].

Hydrogen sorption in a metal hydride reactor occurs over a pressure range determined by the limiting concentrations of the dilute α-phase (hydrogen-free metal) and the concentrated β-phase (saturated metal hydride bed). For instance, in the case of $LaNi_5H_n$, the material predominantly studied in this work, n can vary within the range $[0, 6.7]$. At a temperature of 300 K, the limiting concentrations are approximately $n \approx 0.5$ for the α-phase and $n \approx 6.0$ for the β-phase [57]. Ideally, phase transformation follows a constant-pressure plateau in the pressure-composition isotherm, as governed by the Gibbs Phase Rule. However, actual metal hydrides exhibit plateau slopes and pressure hysteresis.

Fig. 3 illustrates the PCT behavior and van't Hoff plots for metal hydrides, highlighting key thermodynamic and kinetic features, including absorption and desorption plateaus, hysteresis, and phase transitions. The PCT diagram demonstrates the α-phase, β-phase, and the coexistence region (α+β), while the van't Hoff plot shows the enthalpy changes (ΔH) during these transformations. Fig. 3(a) highlights the plateau pressures during absorption and desorption, plateau slopes, hysteresis effects, reversible capacity, and maximum capacity, providing a comprehensive view of the thermodynamic and kinetic characteristics of hydrogen-metal interactions. Similarly, Fig. 3(b) shows the PCT diagram and van't Hoff plot for metal hydrides, depicting the relationship between pressure, temperature, and hydrogen concentration. The van't Hoff plot, plotting ln(P) versus 1/T, relates the thermodynamic relationship with enthalpy change (ΔH).

Recent studies have extensively explored these behaviors in detail; therefore, this work will not delve into them further. The readers are referred to comprehensive articles such as [65,68,73] for further insights. Ge [73] provides a detailed perspective of these phenomena and highlights how hysteresis results in a difference between the absorption and desorption pressures, while the plateau slope varies with temperature, influencing both hydrogen storage capacity and phase transition dynamics. The enthalpy change associated with phase transformation is either released or absorbed at the phase boundaries, whereas heat exchange is negligible within the pure α- or β-phase regions [57]. Additional insights into these phenomena are detailed in Fukai's book [65].

In modeling metal–hydride hydrogen storage systems, conservation of mass, momentum, and energy is always treated as a coupled set of partial differential equations, with the hydride reaction kinetics incorporated as empirical rate expressions in the source terms. To maintain alignment with the standard presentation in the literature and to preserve the clarity of how these equations are coupled, the discussion in this section is organized by the primary physical variable. Within each subsection, variations in modeling assumptions—such as the adoption of local thermal equilibrium (LTE) or local thermal non-equilibrium (LTNE), pore-scale closures, and other important details are highlighted to enable comparative understanding of different methodological choices.

2.2.2. Mass and momentum conservation

The mathematical model for heat, mass, and momentum transfer integrates the principles of mass, energy, and momentum conservations within the porous MH bed. These governing principles are coupled to derive the governing equations for hydrogen storage reactors [33,60,

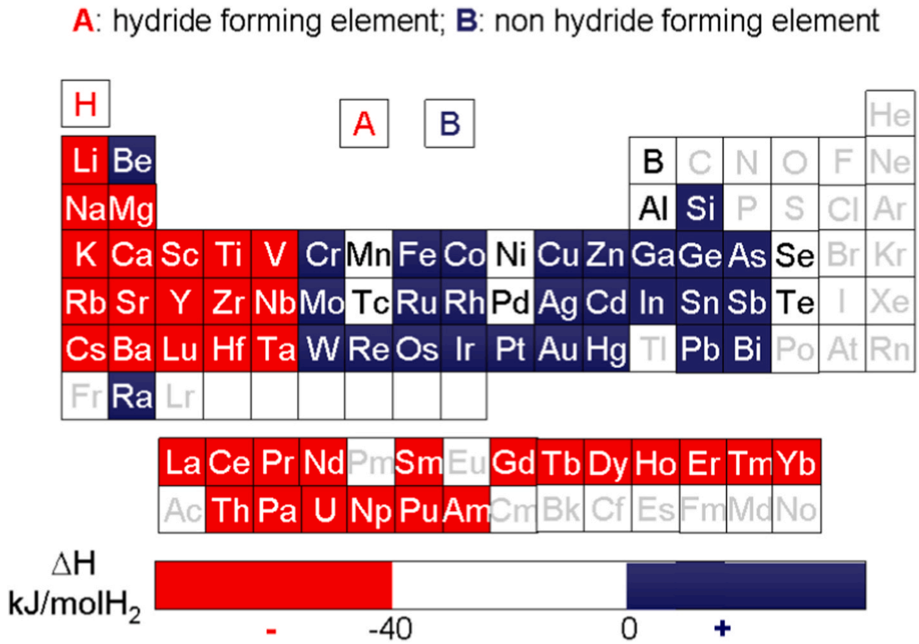

Fig. 2. Periodic table representation highlighting hydride-forming (A) and non-hydride-forming (B) elements (Reprinted from Dornheim's work [67]).

74]. The mass conservation equations for the gaseous and solid phases are represented by Eqs. (6)–(8) respectively [60,75,76].

$$\frac{\partial \varepsilon_{MH}\rho_g}{\partial t}+\nabla\left(\rho_g\vec{U}\right)=-\dot{m} \tag{6}$$

In Eq. (6), $\vec{U}$ represents the velocity field describing the gas movements within the bed driven by the pressure gradients. Assuming hydrogen behaves as an ideal gas (as per Eq. (7)) and incorporating Darcy's law (Eq. (8)), the mass conservation equation for hydrogen is expressed in Eq. (9), where ρ_g represents the density of hydrogen in the bed. The parameters K, and μ represent the bed permeability and dynamic viscosity of hydrogen respectively, while ∇P_g denotes the pressure gradients driving the mass flow in the porous MH bed.

Nasrallah et al. [29] modified Eq. (9) to model heat and mass transfer in a two-dimensional cylindrical domain by applying differentiation rules, leading to the formulation of Eq. (10).

$$\rho_g=\frac{M_gP_g}{RT_g} \tag{7}$$

$$\vec{U}=-\frac{K}{\mu}\nabla P_g \tag{8}$$

$$\frac{\varepsilon M_g}{R}\frac{\partial\left(\frac{P_g}{T_g}\right)}{\partial t}-\frac{K}{\vec{U}}\nabla^2P_g=-\dot{m} \tag{9}$$

$$\frac{\varepsilon M_g}{R}\frac{1}{T_g}\frac{\partial(P_g)}{\partial t}+\frac{\varepsilon M_gP_g}{R}\frac{\partial}{\partial t}\left(\frac{1}{T_g}\right)-\frac{K}{\vec{U}}\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial(P_g)}{\partial r}\right)-\frac{K}{\vec{U}}\frac{\partial^2(P_g)}{\partial z^2}=-\dot{m} \tag{10}$$

In many studies the pressure gradients are considered negligible, due to the low hydrogen velocity [33,77,78]. A detailed investigation on the effects of incorporating pressure gradients and hydrogen velocity will be discussed in the later section of the paper. Under the assumption of uniform pressure, the mass transfer equations for gaseous and solid phase are given by Ref. [79]:

$$\frac{\partial \varepsilon_{MH}\rho_g}{\partial t}=-\dot{m} \tag{11}$$

$$\frac{\partial(1-\varepsilon_{MH})\rho_s}{\partial t}=\dot{m} \tag{12}$$

In modeling metal hydride (MH) reactors, it is important to account for momentum transport in addition to heat and mass transfer. To determine $\vec{U}$, correlations for gas velocity and pressure distribution in packed beds are commonly employed. While Darcy's law given by Eq. (3) is extensively used in MH research and provides an empirical description of fluid movement through granular media [5,7,80], many studies in the literature model it based on the Navier-Stokes equation [20,36,81]. Nam et al. [75] considered Eq. (13) derived from Navier-Stokes equation and took into account the effects of buoyancy. On the other hand, Mohammadshahi et al. [7] presented Eq. (15) and Eq. (16) to model gas movements in the radial (r) and axial (z) directions neglecting the buoyancy term. S_u is the source term in these equations and is expressed in Eq. (17) as a function of the permeability, K, and dynamic viscosity, μ.

$$\frac{1}{\varepsilon}\left[\frac{\partial\rho^g\vec{U}}{\partial t}+\frac{1}{\varepsilon}\nabla\cdot\left(\rho^g\vec{U}\vec{U}\right)\right]=-\nabla P+\nabla\cdot\tau+S_u+\rho^g\vec{g} \tag{14}$$

$$\rho\frac{\partial u_r}{\partial t}=-\frac{\partial P}{\partial r}+\mu\left(\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(ru_r)\right)+\frac{\partial^2u_r}{\partial z^2}\right)-\rho\left(u_r\frac{\partial u_r}{\partial r}+u_z\frac{\partial u_r}{\partial z}\right)+S_u \tag{15}$$

$$\rho\frac{\partial u_z}{\partial t}=-\frac{\partial P}{\partial z}+\mu\left(\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(ru_z)\right)+\frac{\partial^2u_z}{\partial z^2}\right)-\rho\left(u_r\frac{\partial u_z}{\partial r}+u_z\frac{\partial u_z}{\partial z}\right)+S_u \tag{16}$$

$$S_u=-\left(\frac{\mu}{K}\right)\vec{U} \tag{17}$$

The permeability K is often calculated using the Kozeny-Carman equation (Eq. (18)) and the relationship between mean particle diameter and porosity of the bed can be expressed by Eq. (19) [82], where d_p denotes the mean particle diameter and a_s represents the specific surface area of the bed.

$$K=\frac{1}{5}\frac{\varepsilon_{MH}^3}{(1-\varepsilon_{MH})^2}\frac{1}{\left(\frac{6}{d_p}\right)^2} \tag{18}$$

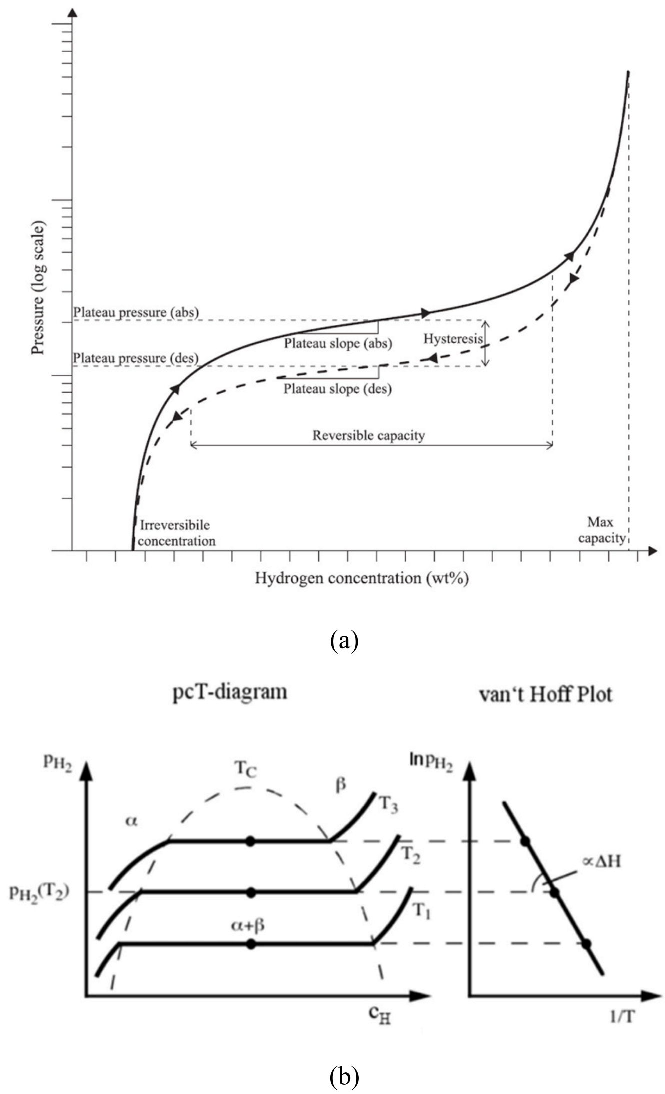

Fig. 3. (a) Schematic representation of Pressure-Concentration-Temperature (PCT) behavior for metal hydrides during absorption (solid line) and desorption (dashed line) (Reprinted with permission from Ref. [68]) (b) Illustration of a PCT diagram and van't Hoff plot for metal hydrides, demonstrating the relationship between pressure, temperature, and hydrogen concentration (Reprinted from Dornheim's work [67]).

$$d_p = \frac{6(1-\varepsilon_{MH})}{a_s} \tag{19}$$

The Blake-Kozeny equation given by Eq. (20) is another important correlation used to describe fluid flow in porous media [25,27]:

$$\vec{U} = -\frac{d_p^2}{150\mu}\frac{\varepsilon_{MH}^3}{(1-\varepsilon_{MH})^2}(\nabla P) \tag{20}$$

This equation models the packed bed as a collection of tubes, modifying the hydrodynamic calculations of single straight tubes to account for pressure drop due to fluid passage through a uniform cross-section conduit [82].

The porosity of the MH bed (ε_{MH}) can be defined based on Eq. (21) [7], where ρ_{solid} refers to the density of bed material in solid form and ρ_{powder} stands for the effective density of the powder. The effective density of the powder is dependent on the compactness of the powder and varies directly with the degree of compression. For lanthanum Nickle, $LaNi_5$, the porosity is considered to be 0.5 in most studies while some studies calculated it to be 0.63 [7,83].

$$\epsilon_{MH} = 1 - \frac{\rho_{powder}}{\rho_{solid}} \tag{21}$$

While considered constant in almost all modeling studies, porosity is actually variable and may change due to the expansion/compression of the bed during absorption/desorption reactions [84,85]. Wijayanta et al. [84] proposed Eq. (22) to account for these changes in porosity as a function of hydrogen content in the bed, where E_p and ϵ_0 represent the expansion coefficient and initial porosity of the bed. Conversely, Minko et al. [85] considered the changes in the bed structure while assuming a constant volume of the MH tank and presented an alternative equation (Eq. (23)) to model the changing porosity.

$$\epsilon_{MH} = 1 - \left(1 - E_p X\right)(1-\epsilon_0) \tag{22}$$

$$\epsilon_{MH} = 1 - \frac{(1-\epsilon_0)(\rho_{emp})}{\rho_{emp} + (\rho_{sat} - \rho_{emp})X} \tag{23}$$

Abdin et al. [57] considered the changes in the average porosity of the bed (ϵ_{MH}) due to the expansion and contraction of individual MH particles during absorption and desorption. They defined ϵ_{MH} in terms of the total volume of solid particles V_s, and the total volume (both particles and pores) of the bed V_{tot} as shown in Eq. (24).

$$\epsilon_{MH} = 1 - \frac{V_s}{V_{tot}} \tag{24}$$

The variation in average porosity is influenced by the relative expansion factor τ_p that is approximately 0.243 for $LaNi_5$ bed [57]. The overall relative change in total pore volume is not simply $1-\tau_p$ because mechanical compaction between expanding particles can increase the packing fraction, offsetting the expected change [57]. Matsushita et al. [86] provided a formulation to calculate the resulting porosity relative to the initial porosity ϵ_0 under absorption and desorption conditions:

FOR ABSORPTION:

$$\epsilon_{MH} = 1 - (1-\epsilon_0)\left(\frac{1+\tau_p \cdot X}{1+\tau_a \cdot X}\right) \tag{25}$$

FOR DESORPTION:

$$\epsilon_{MH} = \begin{cases} 1-(1-\epsilon_0)\left(\dfrac{1+\tau_p \cdot X}{1+\tau_a \cdot X_{max} - \left(1-\dfrac{\tau_p}{\tau_a}X\right)\tau_d}\right) & , \text{if } X \in \left(0, \dfrac{\tau_a}{\tau_p}\right) \\ 1-(1-\epsilon_0)\left(\dfrac{1+\tau_p \cdot X}{1+\tau_a \cdot X_{max}}\right) & , \text{if } X \in \left(\dfrac{\tau_a}{\tau_p}, 1\right) \end{cases} \tag{26}$$

Here, τ_a and τ_d denote the packed bed expansion and contraction ratios during absorption and desorption reactions, respectively. Eqs. (25) and (26) account for the effects of particle expansion, compaction, and mechanical interactions within the bed.

2.2.3. Reaction kinetics

The source term ($\dot{m}$) is obtained from the reaction kinetics and represents the absorption/desorption reaction rate of hydrogen per unit volume at a specific operating pressure given by Eq. (27) and Eq. (28) [77,87,88]. In other words, it relates the hydrogen uptake/release rate with the species concentrations [7]. To establish a relationship for an MH reactor, Supper et al. [89] and Suda et al. [90] conducted experiments for the hydrogenation of metals like $LaNi_5$ and $MmNi_5$. The reaction kinetics relations for the heat and mass transfer in AB_5 types of MH systems presented by Mayer et al. [17] have served as a foundation for subsequent researchers in modeling MH reactors [19,22,23,29].

$$\dot{m}_a = k_a \cdot e^{\left(-\frac{E_a}{R\,T_b}\right)} \cdot \ln\left(\frac{P_g}{P_{eq,a}}\right) \cdot (\rho_{sat} - \rho_s) \tag{27}$$

$$\dot{m}_d = k_d \cdot e^{\left(-\frac{E_d}{R\,T_b}\right)} \cdot \left(\frac{P_g - P_{eq,d}}{P_{eq,d}}\right) \tag{28}$$

In Eq. (4a) and Eq. (4b), ρ_{sat} and ρ_s represent the saturation, and the instantaneous hydride bed densities respectively. k_a and k_d are the rate constants, while E_a and E_d represent the activation energy for hydrogen absorption and desorption reactions respectively.

The equilibrium pressures, $P_{eq,a}$ and $P_{eq,d}$ play a significant role in controlling the rates of both absorption and desorption reactions, as evident from Eqs. (27) and (28). The difference between the equilibrium pressure and the hydrogen pressure in the domain serves as the driving force for sorption processes. This implies that lower equilibrium pressure, given the same hydrogen pressure, leads to faster hydrogen absorption, while a higher equilibrium pressure enhances desorption [7].

The van 't Hoff equation, shown by Eq. (29), has been used in the literature to relate the equilibrium pressure (P_{eq}) to the absolute bed temperature (T_b) and the changes in enthalpy (ΔH) and entropy (ΔS) at a specific reacted fraction (X) [75,91].

$$P_{eq} = \exp\left(\frac{\Delta H}{RT_b} - \frac{\Delta S}{R_g}\right) \tag{29}$$

For a specific metal hydride, the constant value of the change in enthalpy and entropy at a particular reaction fraction implies that the equilibrium pressure is not solely a function of bed temperature but also depends on the reacted fraction (H/M atomic ratio) [64,92]. The correlation of the equilibrium pressure with H/M atomic ratio and bed temperature, derived through van 't Hoff equation, is given by Eq. (30).

$$P_{eq} = f\left(\frac{H}{M}\right) \cdot \exp\left(\frac{\Delta H}{R}\left(\frac{1}{T_b} - \frac{1}{T_{ref}}\right)\right) \tag{30}$$

In Eq. (30), $f\left(\frac{H}{M}\right)$ is a function of the H/M atomic ratio and represents the equilibrium pressure at the constant temperature T_{ref}. Various representations of this function are deduced from the fitting of experimental data in the literature [7]. Eqs. (31) and (32) represent the functions used by Nam et al. [75] and Jemni and Nasrallah [23] respectively.

$$P_{eq} = \left(a_0 + \sum_{n=1}^{7} a_n \left(\frac{H}{M}\right)^n\right) \cdot \exp\left(\frac{\Delta H}{R_g}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right) \tag{31}$$

Where:

$a_0 = -0.34863$ $a_1 = 10.1059$ $a_2 = -14.2442$ $a_3 = 10.3535$

$a_4 = -4.20646$ $a_5 = 0.962371$ $a_6 = -0.115468$ $a_7 = 0.00563776$

$$P_{eq} = 84.811\left(\frac{H}{M}\right) - 402.122\left(\frac{H}{M}\right)^2 + 908.541\left(\frac{H}{M}\right)^3 - 972.495\left(\frac{H}{M}\right)^4 + 395.754\left(\frac{H}{M}\right)^5 \times \exp\left[-3593.82\left(\frac{1}{T_g} - \frac{1}{333}\right)\right] \times 10^5 \tag{32}$$

Many other studies in the literature use an alternative approach based on the reacted fraction to solve the rate of absorption/desorption per unit volume [50,64,93]. The formulation for that approach is presented in Eq. (33)-Eq. (36). Following a linear interpolation and taking $\rho_s = \rho_{sat}$ when $X = 1$, and $\rho_s = \rho_{emp}$ when $X = 0$, the equation representing the reacted fraction X is shown by Eq. (33). Here ρ_{emp} stands for the empirical density representing the hydrogen-free metal bed density.

$$X = \frac{\rho_s - \rho_{emp}}{\rho_{sat} - \rho_{emp}} \tag{33}$$

$$\dot{m} = \frac{(1-\varepsilon_{MH})\rho_{MH}}{M_{MH}}\left[\frac{H}{M}\right]_{max}\frac{dX}{dt} \tag{34}$$

$$\frac{dX}{dt} = k_a \cdot e^{\left(-\frac{E_a}{R\,T_b}\right)} \cdot \ln\left(\frac{P_g}{P_{eq,a}}\right)(1-X) \tag{35}$$

$$\frac{dX}{dt} = k_d \cdot e^{\left(-\frac{E_d}{R\,T_b}\right)} \cdot \left(\frac{P_g - P_{eq,d}}{P_{eq,d}}\right)(X) \tag{36}$$

Chaise et al. [94] and Ye et al. [50] presented the implicit set of reaction kinetic equations for sorption process in the magnesium hydride reactor using the Arrhenius law as shown in Eqs. (37)–(39).

$$\frac{dX}{dt} = k_a \exp\left(-\frac{E_a}{R \cdot T_b}\right) \cdot \left(\frac{P_g}{P_{eq}} - 1\right) \cdot \frac{X-1}{2\ln(1-X)}, \text{ for } P_g > 2P_{eq} \tag{37}$$

$$\frac{dX}{dt} = k_a \exp\left(-\frac{E_a}{R \cdot T_b}\right) \cdot \left(\frac{P_g}{P_{eq}} - 1\right) \cdot (1-X), \text{ for } P_{eq} < P_g < 2P_{eq} \tag{38}$$

$$\frac{dX}{dt} = k_d \exp\left(-\frac{E_d}{R \cdot T_b}\right) \cdot \ln\left(\frac{P_{eq}}{P_g}\right) \cdot \frac{2x}{(-\ln X)^{0.5}}, \text{ for } P_g < P_{eq} \tag{39}$$

Achieving the most appropriate equilibrium pressure (P_{eq}) is crucial for efficient MH tank design, emphasizing the need for accurate expressions for P_{eq} [84,95]. While a temperature dependent van 't Hoff relation is widely used by researchers to determine the equilibrium pressures, there are many studies utilizing the PCT and reaction kinetics parameters for determining the equilibrium pressures [60,96]. Eq. (40) and Eq. (41) show the relations for absorption and desorption equilibrium pressures, respectively. These equations consider the effects of hysteresis and plateau slope [96] and exhibit a strong dependency on both the bed temperature and the reacted fraction. The values for the PCT parameters for absorption and desorption changes slightly in different studies as can be noticed from the studies done by Hasnain et al. and by Yang et al. [60,76].

$$P_{eq,a} = \exp\left(\frac{\Delta S}{R} - \frac{\Delta H}{R\,T_b} + (\phi + \phi_0) \cdot \tan\left(\pi\left(X - \frac{1}{2}\right)\right) + \frac{\beta}{2}\right) \tag{40}$$

$$P_{eq,d} = \exp\left(\frac{\Delta S}{R} - \frac{\Delta H}{R\,T_b} + (\phi - \phi_0) \cdot \tan\left(\pi\left(X - \frac{1}{2}\right)\right) - \frac{\beta}{2}\right) \tag{41}$$

Yao et al. [79], Elkhatib et al. [33] and many other researchers [13,33,36–38,43] used a different approach by assigning $\frac{\Delta S}{R} - \ln(10) = A$ and $\frac{\Delta H}{R} = B$ and considered the simplified version of the equilibrium pressure equation, obtained from Van't Hoff equation, that is only a function of temperature as listed in Eq. (42).

$$\ln\left(\frac{P_{eq}}{P_{ref}}\right) = A - \frac{B}{T_b} \tag{42}$$

Where $P_{ref} = 10$ bars.

In Eq. (42) the values of A and B were considered to be 10.7 and 3704.6, respectively, and the reference pressure P_{ref} was taken as 10 bars. Elkhatib et al. [33] utilized the same relation for the equilibrium pressure but they considered $A = 14.045$ and $B = 3719$ based on the experimental data, followed by regression analysis. Nam et al. [75] conducted a three-dimensional modeling and simulation study focused on hydrogen absorption and performed a comparative evaluation of the two methods for determining equilibrium pressures. The study revealed distinct hydrogen absorption characteristics for each approach. It was determined that the equilibrium pressure is highly dependent on both the H/M atomic ratio and the temperature of the hydride bed.

2.2.3.1. Effects of pressure gradients in the bed. To assess the impact of hydrogen flow in MH tanks, Chaise et al. [94] introduced two dimensionless numbers, C and N, to evaluate the effect of pressure gradients on the hydrogen absorption process. Using the developed criteria, they demonstrated that for a combination of microstructural and material properties the pressure gradients, as well as the convective heat transfer in the bed, can be neglected. Recently, Hasnain et al. [5,61] investigated the effect of Darcy's velocity due to pressure gradients within the MH bed on the heat and mass transfer model. It was concluded that incorporating the pressure gradients using Darcy's law introduced increased stiffness and numerical instability into the coupled equations, however, their effect on model outcomes was negligible. On the other hand, the study conducted by Freni et al. [35] suggest that changing the pressure gradients due to Darcy's velocity can have a prominent impact on the performance of MH reactors.

The dimensionless numbers presented by Chaise et al. [94], to define a criterion for assessing the impact of pressure gradients in the bed, are as follows:

If $C \gg 1$ and $N \ll 1$, the hydrogen pressure in the tank is considered uniform, where C and N are the dimensionless parameters given by Eqs. (43) and (44) respectively.

$$C = \frac{\partial P_{eq}}{\partial T} \cdot \frac{r_4^2 \cdot (1-\varepsilon) \cdot wt \cdot \rho_m \cdot \Delta H}{\lambda_e M} \cdot k_a \cdot \frac{e^{-Ea/RT}}{P_{eq}} \tag{43}$$

$$N = \frac{\lambda_e \cdot M \cdot d^2 \cdot \mu_g}{\frac{\partial P_{eq}}{\partial T} \cdot \Delta H \cdot \rho_g \cdot K \cdot r_4^2} \tag{44}$$

2.2.4. Energy conservation

2.2.4.1. Local thermal NON-EQUILIBRIUM (LTNE) model. Considering local thermal non-equilibrium within the MH bed, the energy balance equations can be presented separately for the gas and solid phases in the porous MH bed. In 1995 Jemni and Nasrallah presented their two-dimensional heat and mass transfer models for absorption and desorption in the MH bed [23,97] proposing the following LTNE equations for each phase, while neglecting the pressure gradients in the bed.

Solid phase

$$(1-\varepsilon)\rho_s^s C_{ps} \frac{\partial T_s}{\partial t} = (1-\varepsilon)\lambda_s \nabla^2 T_s - H_{gs}(T_g - T_s)A - \dot{m}\,(\Delta H^{\circ} + C_{ps} T_s - C_{pg} T_g) \tag{45}$$

Gas phase

$$\varepsilon \rho_g^g C_{pg} \frac{\partial T_g}{\partial t} = \varepsilon \lambda_g \nabla^2 T_g - \rho_g^g C_{pg} \vec{U} \cdot \nabla T_g + H_{gs}(T_g - T_s)A - \dot{m}\, C_{pg}(T_g - T_s) \tag{46}$$

In 1999, Jemni and Nasrallah [83] conducted a theoretical and experimental investigation of a metal-hydrogen reactor. They introduced a slightly modified heat transfer model for the gas phase within the porous MH bed given by Eq. (47).

$$\varepsilon \rho_g^g C_{pg} \frac{\partial T_g}{\partial t} = \varepsilon \lambda_g \nabla^2 T_g - \rho_g^g C_{pg} \vec{U} \cdot \nabla T_g - H_{gs}(T_g - T_s)A - \max(-\dot{m}, 0) C_{pg}(T_g - T_s) \tag{47}$$

In Eqs. (45)–(47) "s" and "g" stand for the solid and gas phases, A represents the solid-gas exchange area, and λ is the thermal conductivity. $\dot{m}$ is obtained from the reaction kinetics [75,77,87,88] and H_{gs} signifies the interphase heat transfer coefficient between the gas and solid phases.

The heat transfer mechanisms for both phases encompass heat conduction $(\lambda \nabla^2 T)$, heat convection between gas and solid phases $\left(H_{gs}\left(T_g^g - T_s^s\right)S\right)$, and changes in molecular energy associated with sorption reactions $\left(\dot{m} C_{pg}\left(T_g^g - T_s^s\right)\right)$. Additionally, Eqs. (46) and (47) consider heat transfer due to gas movement $\left(\rho_g^g C_{pg} \vec{V} \cdot \nabla T_g^g\right)$, and the enthalpy changes of the hydriding and dehydriding reactions $(\dot{m}\,\Delta H)$ are also incorporated.

Several empirical equations have been formulated to express the interphase heat transfer coefficient, establishing relationships between the Nusselt (Nu) and Reynolds (Re) and Prandtl (Pr) numbers [98]. Among these, the two frequently employed correlations for packed beds [98,99] are the ones proposed by Gunn [100] and Saez and McCoy [101] as given by Eq. (48) and Eq. (49) respectively:

$$H_{gs} = \frac{\lambda_g}{d_p}\left[(7 - 10\varepsilon + 5\varepsilon^2)(1 + 0.7 Re^{0.2}\ Pr^{1/3}) + (1.33 - 2.40\varepsilon + 1.20\varepsilon^2) Re^{0.7}\ Pr^{1/3}\right] \tag{48}$$

$$H_{gs} = \frac{\lambda_g}{d_p}(2 + 1.1 Re^{0.6}\ Pr^{1/3}) \tag{49}$$

Here, d_p represents the average MH particle diameter, which for $LaNi_5$ was reported to be decreasing from 150 μm to 10 μm during pulverization [84,102]. Jemni and Nasrallah [23] in their absorption model utilized Eq. (49) for determining the heat transfer coefficient between both phases in the porous bed.

2.2.4.2. Local thermal equilibrium (LTE) model. The local thermal equilibrium hypothesis has been a common consideration in nearly all theoretical studies of MH systems. This assumption implies that the temperature of the gas and solid storage bed remains uniform throughout the entire bed volume.

While prior MH studies frequently validate LTE (e.g., interphase/near-wall differences $\lesssim 1$ K in typical regimes), its validity has been assessed in detail by Jemni and Nasrallah [23,29]. They showed that the difference between T_s obtained from Eq. (45) compared with the calculated T_b from Eq. (50) is negligible, as is the difference between T_s and T_g. Only near cooling walls did deviations slightly exceed ~ 1 K thus the LTE assumption was generally found to hold. Consistent with this, Quintard and Whitaker [103,104] investigated and supported the local thermal equilibrium hypothesis for moving fluids in solid beds, and Hasnain et al. [5] demonstrated that the LTE formulation can be derived from the limit of the LTNE framework (derivation provided in their supplementary material). Fig. 4 depicts the heat transfer process between the solid particle and gaseous region which can be analyzed based on the Biot number (Bi) criteria, distinguishing between Local Thermal Equilibrium (LTE) for $Bi \ll 1$ and Local Thermal Non-Equilibrium (LTNE) for $Bi \gg 1$.

Nevertheless, deviations for LTE arise under steep transients and close to cooling walls as found in studies outside the area of metal hy-

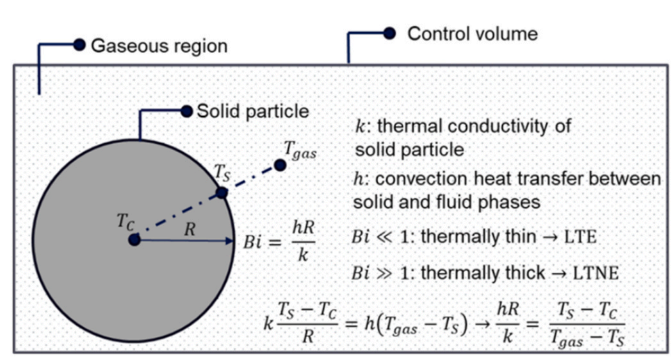

Fig. 4. Schematic of heat transfer between a solid particle and gaseous region, highlighting the control volume and Biot number (**Bi**) criteria, with **Bi≪1** indicating Local Thermal Equilibrium (LTE) and **Bi≫1** indicating Local Thermal Non-Equilibrium (LTNE), governed by the heat balance equation.

dride modeling [105,106]. Moreover, it should also be noted that the LTNE model requires specification of the convective heat transfer coefficient between the gas and solid phases, as well as the emissivity of the solid phase to account for radiative exchange. However, introducing these additional terms increases the uncertainty of the model, since accurately determining these parameters is particularly challenging. Beyond these case-specific findings, recent studies propose decision criteria rather than universal percentages. For example:

1. In aquifer and sediment systems, Shi et al. [105] formulate two simple criteria $g(d_p, u)$ and $f(d_p, u)$ (functions of grain size d_p and Darcian velocity u) to flag the onset and significance of LTNE; they show that LTNE effects are expected for $d_p \geq 10$ mm at $u \leq 2\ m\ d^{-1}$ (meters per day), and that using LTE in such regimes can severely bias effective thermal diffusivity and inferred velocities (sometimes by orders of magnitude).
2. For forced convection, "new criterion" formulations relate LTE acceptability to the interfacial Biot number, solid–fluid conductivity ratio, pore length scale, porosity, and imposed heat flux. LTE is reliable when interfacial exchange is strong and property contrasts are modest, but degrades as interfacial resistance grows or contrasts increase [107].
3. Exact LTNE solutions for porous media further reveal temperature-gradient bifurcation and marked wall-Nusselt deviations when internal heat generation is present, underscoring that LTNE is required in such operating windows [108].
4. Laboratory-scale evidence further strengthens these findings: Lee et al. [109] conducted controlled heat transport experiments in porous media with grain sizes ranging from 5 to 30 mm and Darcy velocities between 3 and 23 $m\ d^{-1}$. They observed that for small grains (5 – 15 mm), LTE and LTNE models produced nearly identical predictions (RMSE differences < 0.01), whereas for larger grains (20 – 30 mm) and high velocities $\left(\geq 17\ m\ d^{-1}\right)$, LTNE diverges from LTE, with fluid–solid temperature differences exceeding 5% of the system's temperature gradient.

These findings confirm that LTE breaks down in coarse-grained, fast-flowing porous systems, consistent with theoretical criteria. The improved fidelity comes with higher computational cost because LTNE solves two coupled energy equations with interfacial exchange. In practice, LTNE adoption in industry is limited due to (i) the need to identify/calibrate interfacial heat-transfer coefficients and phase-specific properties and (ii) longer runtimes and solver robustness [7, 110]. Presently, many metal hydride systems fall within the range where LTE models are sufficient, but the aforementioned criteria can be used to predict the limits at which LTNE models will become necessary.

By applying the LTE assumption, Eqs. (45)–(47) can be combined into a single equation for the energy balance. Additional considerations are applied to derive different versions of this equation. Eq. (50) accounts for the impact of Darcy's velocity due to pressure gradients [7,60, 75], while Eqs. (51) and (52) neglects the pressure gradients. Eq. (51) completely neglects the convection heat transfer between the two phases [32,111] while Eq. (52) incorporates an additional source term in the equation to account for the heat exchange due to sorption process [33, 78,83].

$$\frac{\partial\left(\rho_b C_{p,b}\right)_{eff} T_b}{\partial t} + \nabla\cdot\left(\rho_g C_{p,g} \vec{U} T_b\right) = \nabla\left(\lambda_{eff}\nabla T_b\right) + \dot{m}\cdot\Delta H \tag{50}$$

$$\frac{\partial\left(\rho_b C_{p,b}\right)_{eff} T_b}{\partial t} = \nabla\left(\lambda_{eff}\nabla T_b\right) + \dot{m}\cdot\Delta H \tag{51}$$

$$\frac{\partial\left(\rho_b C_{p,b}\right)_{eff} T_b}{\partial t} = \nabla\left(\lambda_{eff}\nabla T_b\right) + \dot{m}\cdot\Delta H + \dot{m}\cdot T_b\left(C_{pg} - C_{ps}\right) \tag{52}$$

In some studies, the transient effect of pressure variation on the heat transfer is explicitly considered and incorporated in the energy balance equation for the metal hydride reactors as shown in Eq. (53) [27,112, 113].

$$\frac{\partial\left(\rho_b C_{p,b}\right)_{eff} T_b}{\partial t} + \nabla\cdot\left(\rho_g C_{p,g} \vec{U} T_b\right) = \nabla\left(\lambda_{eff}\nabla T_b\right) + \dot{m}\cdot\Delta H + \epsilon_{MH}\vec{U}.\nabla P_g + \frac{\partial P_g}{\partial t} \tag{53}$$

The coefficients of the transient term in the energy equation, representing the effective bulk heat capacity is expressed as the sum of the products of the volume fraction, density, and specific heat for each phase within the system, as given by Eq. (54) [60], where the summation accounts for the contributions from all constituent phases in the bed. Similarly, the term λ_{eff}, representing the effective thermal conductivity of the bed, is given by Eq. (55) [5].

$$\left(\rho_b C_{p,b}\right)_{eff} = \sum_{i=1} \varepsilon_i \rho_i C_{pi} = \varepsilon_{MH} \rho_g C_{p,g} + (1-\varepsilon_{MH})\rho_s C_{p,s} \tag{54}$$

$$\lambda_{eff} = \left[\varepsilon_{MH}\lambda_g + (1-\varepsilon_{MH})\lambda_{MH}\right] \tag{55}$$

Instead of considering the variations in the thermal conductivity due to microstructural properties and porosity of the MH bed, some studies have used a constant value. For instance, Elkhatib et al. adopted an effective thermal conductivity of 1.28 $W(mK)^{-1}$ [33], while several other studies have used a value of 1.087 $W(mK)^{-1}$ for the MH bed thermal conductivity [114,115].

Thermophysical properties critical to MH modeling vary across studies. For example, Nam et al. [75] provided detailed specific heat and thermal conductivity values for $LaNi_5$, emphasizing their role in accurate heat transfer predictions. Wijayanta et al. [84] proposed equations to account for porosity changes during absorption/desorption. Other studies, such as Abdin et al. [57], quantified bed permeability to refine momentum transport calculations.

Table 3 presents the percent weight capacity and thermal conductivities of some commonly used metal hydrides under different operating temperatures and pressures along with modeling recommendations. This provides direct recommendations on whether LTE or LTNE assumptions are appropriate under specific material and operational regimes, thereby linking thermophysical parameters to modeling strategy. Similarly, the parameters governing the reaction kinetics are shown in Table 4.

2.3. Initial and boundary conditions

To model MH reactors effectively, it is important to specify appropriate initial and boundary conditions based on the reactor's shape, operational parameters, and cooling system configuration. This often involves the application of homogeneous or non-homogeneous Neumann boundary conditions and Dirichlet boundary conditions.

In most modeling studies, it is assumed that both the solid particles and hydrogen gas are in thermodynamic equilibrium. This assumption implies that the bed's initial temperature is uniform, typically set to room temperature.

For axis-symmetrical cylindrical MH reactors with external heating or cooling, the homogeneous Neumann's or zero-flux boundary condition, given by Eq. (56) for 2D cylindrical coordinates, can be applied at the axis of symmetry $(r = 0)$.

$$\frac{\partial T}{\partial r}(z, r=0, t) = 0, \frac{\partial P}{\partial r}(z, r=0, t) = 0, \frac{\partial u_r}{\partial r}(z, r=0, t) = 0 \tag{56}$$

Similarly, the same no-flux boundary condition is applicable for any insulated walls in any configuration or shape of the reactor, irrespective of the dimensions of the computational domain.

Table 3
Thermal conductivities for different metal hydrides based on operating conditions with modeling recommendations [1,31,58,116].

MH Material (wt %)	Operating Pressure [MPa]	Operating Temperature [K]	Thermal conductivity of the metal, k_s [W/(m·K)]	Modeling Recommendation
$NaAlH_4$ (5.6 %)	0.1–3	313–453	0.3–1.0	Very low conductivity; requires HTF/PCM augmentation; LTNE preferred under fast cycling due to poor heat transfer
$TiMn_{1.5}H_x$ (1.8 %)	0.1–4	298–333	0.3–1.0	Similar to $NaAlH_4$; LTE acceptable for small ΔT (<5 K) since operating temperature range is lower, but LTNE recommended in reactor-scale simulations
MgH_2 (7.5 %)	0.02–1	523–673	2.5–8.0	Higher conductivity but high operating temperature; LTE sufficient for bulk modeling; LTNE useful only for steep wall gradients
$LaNi_{4.7}Al_{0.3}H_x$ (1.2 %)	0.002–5	253–393	0.1–1.0	Very low conductivity; modeling must include enhancement strategies (fins/foam/PCM); LTNE required near cooling walls
$TiFeH_x$ (1.9 %)	8.0	303	1.5	Moderate conductivity; LTE generally adequate; LTNE only if investigating pore-scale gradients
$Ti_{1.1}CrMnH_x$ (1.8 %)	0.3–20	293–313	0.4–0.8	Low conductivity; LTE valid under uniform heating, but LTNE preferred when modeling rapid refueling cycles
$LaNi_5$ (1.3 %)	0.001–1	293–323	0.2–1.5	Benchmark MH material; LTE validated by Jemni & Nasrallah ($\Delta T \lesssim 1$ K) [23,29], but LTNE necessary to capture near-wall or fast transient phenomena
$LiBH_4/MgH_2$ (9.8 %)	0.3–8	873	0.8–1.5	Mixed system; high operating temperature and low thermal conductivity; LTNE recommended if coupled with PCM or foams

Table 4
Parameters related to reactions kinetics and profiles of pressure, hydrogen concentration and temperature (PCT) from Eq. (8) and Eq. (9) [33,60,75,117].

Parameters	Absorption	Desorption	Unit	Reference
Rate constant, $k_{a,d}$	59.187	9.57	[s^{-1}]	[7,33,60]
Activation energy, $E_{a,d}$	21179	16473	[J /mol H_2]	[7,33,60]
P-C-T parameter, A	13.1		–	[33,60]
P-C-T parameter, B	3700		[K]	[60]
Hysteresis factor, β	0.137		–	[33,60]
Plateau flatness factor, ϕ	0.038		–	[60]
Plateau flatness factor, ϕ_0	0		–	[60]
Reaction enthalpy, ΔH	30100 - 30800		[J/mol]	[13,111] -[33,60]
Saturation density, ρ_{sat}	6520 - 8527		[kg/m^3]	[26–33, 43]
Hydrogen-free metal density, ρ_{emp}	6430 - 8400		[kg/m^3]	[33]

$$\nabla T_b = 0, \nabla P_g = 0 \tag{57}$$

For momentum conservation modeled with Navier–Stokes equations, a zero-slip velocity is assumed at the end walls of the MH tanks. Hence, Dirichlet boundary conditions shown by Eq. (58) are imposed on gas velocity:

$$u_z(z, R, t) = 0, u_r(h', r, t) = 0, u_z(0, r, t) = 0, u_r(0, r, t) = 0 \tag{58}$$

where R is the internal radius, and h' is the height of the reactor tank.

The boundary conditions for the heat transfer walls, where the convective heat transfer occurs between the bed and the cooling/heating fluid, are given by Eq. (59) and Eq. (60), where T_f is the temperature of the fluid, h is the convective heat transfer coefficient (considered to be $1500 W/(m^2.K)$ in many studies) and n stands for the normal vector to the corresponding wall [33,60,75].

$$-\lambda_{eff} \nabla T_b \cdot n = h(T_b - T_f) \tag{59}$$

$$\nabla P_g = 0 \tag{60}$$

At the mass exchange boundary, where absorption of hydrogen happens into the bed, a Danckwerts' boundary condition [118] is considered by Yang et al. [60] as shown by Eq. (61) to ensure the continuous flow rate of hydrogen. However, for the simplified model not taking into account the pressure gradients and Darcy's velocity, this boundary is treated as homogeneous Neumann boundary condition given by Eq. (62).

$$-\lambda_{eff} \nabla T_b \cdot n = \rho_{g,in} \vec{U} C_{p,g}(T_{in} - T_b) \tag{61}$$

$$\nabla T_b = 0 \tag{62}$$

Hydrogen entry into a storage tank can be regulated by either a constant pressure or a constant mass flow rate. Fig. 5 demonstrates similar reaction completion times for both approaches, when the initial tank temperature and interior pressure are kept at 293 K and 0.01 MPa respectively. Under fixed pressure, the tank is rapidly pressurized, leading to high initial absorption rates and a sharp rise in temperature. Conversely, a constant mass flow ensures a steady filling process, resulting in a slower temperature increase with lower peak values. However, without pressure constraints, the tank pressure can rise significantly, reaching approximately 6 MPa in 500 s as shown in Fig. 5 (b). Fixing pressure at the inlet is suitable for evaluating advanced temperature dissipation designs, while using a constant mass flow rate provides a realistic representation of tank performance under operational conditions [58].

3. Thermal management modeling

Given the exothermic and endothermic nature of sorption reactions, effective temperature management is crucial in metal hydride (MH) reactors to prevent overheating during absorption and excessive cooling during desorption. Consequently, it is essential to cool the bed during absorption and heat it during desorption. This thermal regulation significantly impacts the equilibrium pressure, thereby leading to faster hydrogen charging and discharging processes. There are three primary thermal management approaches employed and modeled for MH-based hydrogen storage tanks [55].

i. The utilization of a heat transfer fluid (HTF). In the hydrogen absorption process, discharged heat is absorbed by the HTF, while during desorption, the fluid is externally heated to supply the additional heat required for hydrogen release [93,119].
ii. The utilization of phase change materials (PCM) with hydrogen storage tanks [50]. PCM has the capability to store waste heat generated during hydrogen absorption and release it when required for hydrogen desorption, eliminating the need for heat exchangers. This method enables the practical recycling of waste heat, thereby enhancing the overall efficiency of the system [120].

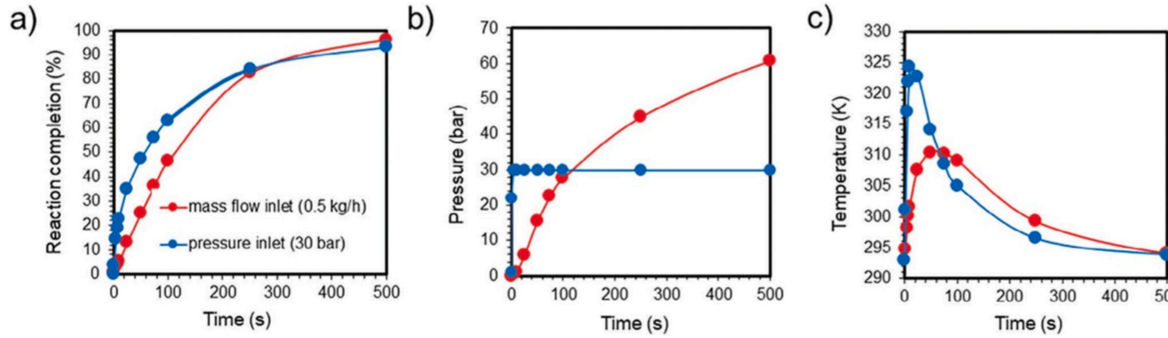

Fig. 5. Comparison of simulation outcomes for an MH tank with fixed hydrogen pressure or mass flow at the inlet: (a) Rate of hydride formation, (b) Pressure variation, and (c) Temperature variation in MH tank (Reprinted from Ref. [58]).

iii. The use of fins to maximize the surface area of the bed with the heat transfer fluid (HTF) or phase change material (PCM), ultimately resulting in an increased heat transfer rate [42].

Limited by the manuscript's emphasis on mathematical formulations and modeling approaches, this work deliberately treats thermal-management techniques (HTF, PCM, fins) at a modeling level rather than attempting an exhaustive analysis of material performance and engineering trade-offs. For readers seeking a detailed synthesis of thermal-management materials and their performance trade-offs, several dedicated recent review articles are available [121–125], which comprehensively benchmark PCM/HTF systems in terms of storage density, thermal response (charge/discharge time), cost, and system-level impacts on gravimetric/volumetric hydrogen storage metrics. We therefore limit the current scope to the representation and coupling of thermal management in MH reactor models and refer interested readers to the cited reviews for systematic material comparisons and design guidance.

Many attempts have been made to optimize the hydrogen storage process in MH hydrogen storage systems by integrating various heat exchangers within hydrogen storage tanks. Common heat exchanger designs explored in the literature include axial heat exchanger tubes, coiled heat exchanger tubes, and transversal heat distribution fins [42]. For instance, Chaise et al. [126] experimentally investigated a metal hydride tank with a central tube heat exchanger and developed a two-dimensional numerical heat transfer model. Similarly, Bao et al. [93] implemented a one-dimensional heat and mass transfer model to analyze temperature, pressure, and density distributions inside magnesium hydride reactors. Kudiiarov et al. [110] performed a comparative analysis of different configurations of heat exchangers, fins, and cooling tubes to identify the most effective design for optimizing reactor performance in terms of heat transfer efficiency and the reactor's weight and size characteristics.

Hydrogen storage materials with diverse characteristics have been developed to suit various applications. Selection of an appropriate material depends on the specific requirements of the application. Manai et al. [127] developed a numerical model for varying configurations of MH bed and demonstrated that incorporating stainless steel fins or compacted Ti-Mn alloy/stainless steel mixtures significantly improved heat transfer, absorption/desorption rates, and storage density for eco-friendly mobility. The alternative configuration also provided a simple and cost-effective recycling option. Liu et al. [58] provides a comprehensive overview of advancements in improving heat management in metal hydride systems. Key strategies include enhancing the thermal conductivity of hydride beds through compaction with binders, metal foams, and optimizing tank designs with cylindrical shapes, fins, and tubular heat exchangers [58,128]. Additionally, integrating phase change materials has shown promise, though it can complicate system designs [58].

Metal foams have emerged as a promising approach for enhancing the thermal conductivity of metal hydride (MH) beds due to their high surface area, porous structure, and excellent thermal conductivity [58,129,130]. Minko et al. [41] conducted a numerical study to evaluate the impact of replacing powder with metal foam and found that while the foam reduced sorption speed due to a minor pressure drop from decreased porosity, it significantly enhanced the reaction rate by improving heat dissipation. Overall, the foam's influence on system performance was complex and could not be attributed to isolated effects.

Among various options, aluminum (Al) foams are the most widely used, as they enable significantly faster hydrogen uptake rates compared to copper (Cu) or zinc (Zn) foams, as illustrated in Fig. 6(b) [58,130] for the $LaNi_5$ reactor design shown in Fig. 6(a). The pore size of the foam also plays a critical role, with smaller pores offering higher heat transfer surface area per unit volume. Fig. 6(c) shows that $LaNi_5$ beds with Al foams having a pore size of 0.125 mm demonstrate faster hydrogen absorption rates than those with 0.2 mm or 0.35 mm pores [130]. Furthermore, increasing the amount of Al foam in the bed enhances hydrogen uptake rates, with studies reporting up to a 65 % improvement in charging time compared to loose-packed powders (Fig. 6(d) [130]. However, challenges remain, such as difficulty in densely packing storage alloys within the foam pores and reduced hydrogen storage capacity due to lose contact between hydride particles and foam walls [58]. A similar observation was made by Laurencelle and Goyette for $LaNi_5$ beds incorporating metal foams with pore sizes of 2.3 mm, 4.6 mm, and 9.2 mm, which correspond to porosities of 40 pores per inch (PPI), 20 PPI, and 10 PPI, respectively [131]. Despite these limitations, metal foams offer a substantial improvement in heat management and hydrogen storage performance, making them a viable option for optimizing MH storage systems.

The optimization study on different configurations of multi-tube heat exchange was done by Bao et al. [132]. Similarly, Ma et al. [133] studied the performance of multi-tubular finned heat exchangers in MH tanks and optimized the heat transfer through fins. The heat exchanger designs are considered under four varying heat transfer configurations including natural convection, finned surfaces, internal circulation of fluid, and with the concentric fins inside the hydride tank [1]. The most efficient design in terms of charging/discharging time is the design with concentric tube for fluid flow with transverse fins attached to the tube [7,134]. These studies show that the improvement is as a result of increased heat transfer areas. However, the complexity of the heat exchangers to further enhance the heat transfer limits the applicability of this approach. Furthermore, the requirement of an external heating source for hydrogen desorption process is another factor contributing towards lower efficiency [50].

Numerous studies have investigated the performance of hydrogen storage tanks with and without the involvement of PCM. Ye et al. [50] develops a two-dimensional numerical model to provide a comparative analysis of the heat transfer mechanisms in the MH-tanks before and after the incorporation of PCM. It was shown that the incorporation of PCM significantly enhanced the tank's performance, improving heat

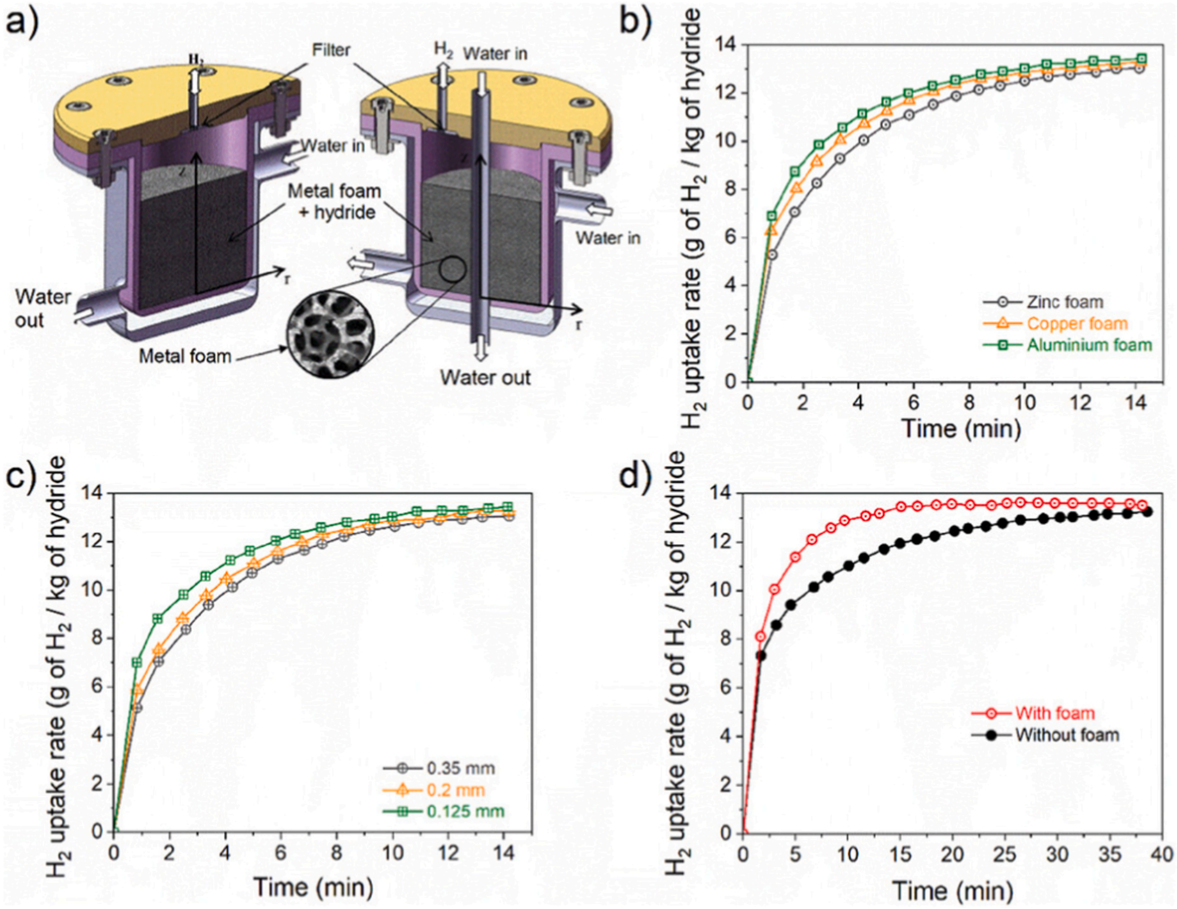

Fig. 6. (a) Schematic of a hydrogen storage system utilizing a $LaNi_5$ bed integrated with metal foam (b) Influence of metal foam type on hydrogen uptake rates (c) Evaluation of hydrogen absorption rates as influenced by variations in metal foam pore size and (d) Comparison of hydrogen absorption rates in hydride beds with and without metal foam integration (Reprinted with permission from Ref. [130] and adopted from Ref. [58]).

transfer and storage efficiency while reducing hydrogen absorption time by 60.2 % compared to a tank without PCM [50]. The effect of fins on the average temperature and conversion rate of the MH bed as well as PCM was also presented in this work. The major results will be presented later in this section.

An experimental study was done by Garrier et al. [135] on magnesium hydride (MgH_2) tanks using $Mg_{69}Zn_{28}Al_3$ as the phase change material. However, the overall gravimetric storage density of the storage system was reduced (to about 0.315 wt%) due to the involvement of heavy-weight PCM alloy. Marty et al. [136] developed a computational model to explain the heat transfer process in tanks with PCM. A modest improvement in the storage efficiency was achieved by using Phase Change Material (PCM) [136]. Satya Sekhar et al. [42] utilized a 3D numerical model to evaluate hydrogen uptake performance in cylindrical metal hydride beds containing $MmNi_{4.6}Al_{0.4}$. They compared four heat exchange configurations: internal cooling with straight or helically coiled tubing, and external cooling without or with transversal fins. Simulation results highlighted the influence of heat transfer coefficient and provided insights into optimizing dynamic performance for each design. A summary of different metal hydride reactors along with their sizes, hydrogen charging rates, and cooling system configurations is provided in Table 5 [58].

Mâad et al. conducted numerical studies to evaluate the influence of the physical properties of various PCMs on absorption and desorption of hydrogen [137,138]. It was found that the utility of PCM, along with many other design and operation variables, depends on the MH material used for storage. $Mg_{69}Zn_{28}Al_3$ was determined to be a suitable PCM for Mg_2Ni hydrogen storage material due to the compatibility of reactor operating temperature range with the melting point of the PCM. Yao et al. [139] performed a numerical study to optimize the storage performance in $LaNi_5$ MH tanks with $Na_2HPO_3{\cdot}7H_2O$ as the PCM. This study investigated various important parameters like the physical properties of PCM and absorption pressure of hydrogen to utilize the maximum storage capacity. Due to the lack of heat exchangers in hydrogen storage tanks with PCM, a huge amount of PCM must be provided to fully absorb the generated heat during absorption and maintain the reaction rate. Resultantly, this highly increases the weight of the tank and significantly reduces the gravimetric hydrogen storage density. Combining these two approaches, a more efficient storage could be designed with both heat exchangers and PCM [50].

3.1. Heat transfer fluid (HTF)

In MH-based hydrogen storage systems, the fluid (that in most cases is water) is forced alongside the bed at a constant velocity to enhance heat exchange. Many different reactors are used in the literature with varying configurations of MH bed and fluid flow. The most common reactor configuration is the one modeled by Jemni and Nasrallah [23,29,97]. The cross-sectional view of this reactor is shown in Fig. 7(a). The second configuration for MH bed and HTF is the one commonly used for modeling hydrides heat pumps and is shown in Fig. 7(b) [60].

Under laminar forced convection, the continuity, momentum, and energy equations for the fluid are given as follows [150]:

$$\nabla \cdot \vec{u} = 0 \tag{63}$$

Table 5
Metal hydride storage tanks configurations and hydrogen charging rates [58].

Storage Material	Mass of the Bed [kg]	Heat Exchanger Design	Hydrogen Charging Rate [kg H_2/min]	Source
$La_{0.95}Ce_{0.05}Ni_5$	50	Hexagonal honeycomb fin structure	0.005	[140]
MgH_2	10	Integrated phase change material	0.029	[141]
$LaNi_5$	5	Embedded cooling tubes with internal conical fins	0.008	[142]
$NaAlH_4$	100	Aluminum fins connecting cooling tubes	0.272	[143]
$LaNi_5$	1	Embedded spiral cooling tube with external water jacket	0.001	[144]
Ti–Mn alloy	210	Multi-tube configuration for hydrogen storage	0.078	[145]
$LaNi_5$	10	External water jacket and cooling tubes	0.004	[146]
$LaNi_5$	–	Reactor with dual coil cooling design	0.001	[147]
$LaNi_5$	–	Reactor with segmented compartments	0.001	[148]
$LaNi_5$	–	Concentric cooling tube with fins	0.003	[134]
$LaNi_{4.25}Al_{0.75}$	0.11	Nitrogen-cooled double-layer reactor	<0.001	[149]

$$\rho_h \frac{\partial \vec{u}}{\partial t} + \rho_h (\vec{u} \cdot \nabla) \vec{u} = -\nabla P + \mu_h \nabla^2 \vec{u} \tag{64}$$

$$\rho_h c_{p,h} \frac{\partial T_h}{\partial t} + \rho_h c_{p,h} \vec{u} \cdot \nabla T_h = \lambda_h \cdot \nabla^2 T_h \tag{65}$$

3.2. Phase change material (PCM)

Yao et al. [79] and Chibani et al. [63] modeled the heat and mass transfer in MH bed by combining it with a simplified PCM heat exchanger as shown in Fig. 8(a). They utilized disodium hydrogen phosphate salt hydrate $Na_2HPO_3{\cdot}7H_2O$ as the phase change material. This compound is the hydrate form of a salt produced by the partial neutralization of phosphoric acid (H_3PO_4). Under the assumption of negligible convective heat transfer within the MH bed [23,77] and PCM [88], and a perfect insulation of the PCM heat exchanger, the following governing equations were presented for the hydrogen storage systems integrated with PCM.

The heat transfer equation for the PCM material involves a transient, diffusion and a constant source term and is given by Eq. (66).

$$\rho_p C_{p,p} \frac{\partial T}{\partial t} = \nabla \cdot (\lambda_p \nabla T) + Q \tag{66}$$

where Q represents the heat supplied into or taken out of the PCM. $C_{p,p}$ and ρ_p are the specific heat capacity and density of PCM respectively. The density and specific heat capacity during sorption process are given by Eq. (67) and Eq. (68) respectively.

$$\rho_p = \theta \rho_{phase\ 1} + (1-\theta) \rho_{phase\ 2} \tag{67}$$

where θ represents the melting fraction of the PCM.

$$C_{p,p} = \frac{1}{\rho_p} \left(\theta \rho_{phase1} C_{p,phase1} + (1-\theta) \rho_{phase\ 2} C_{p,phase2} \right) + L \frac{\partial \alpha_m}{\partial T} \tag{68}$$

Similarly, the effective thermal conductivity (λ_p) of the PCM is dependent on the melting fraction and is given by Eq. (69):

$$\lambda_p = \theta \lambda_{phase1} + (1-\theta) \lambda_{phase\ 2} \tag{69}$$

The parameters α_m in Eq. (70) is given by:

$$\alpha_m = \frac{1}{2} \frac{(1-\theta)\rho_{phase\ 2} - \theta \rho_{phase\ 1}}{\theta \rho_{phase\ 1} + (1-\theta) \rho_{phase\ 2}} \tag{70}$$

3.3. MH reactor with HTF and PCM

Ye et al. [50] developed a model for a magnesium hydride (MH) hydrogen storage system that incorporates both a heat transfer fluid (HTF) and a phase change material (PCM). The HTF used is Dowtherm A synthetic oil, and the MH system consists of several compacted disks, comprising Mg/MgH_2 and expanded graphite (EG), stacked in the tank. The PCM employed is $NaNO_3$. A cross-sectional schematic of the axis-symmetrical MH reactor integrated with both HTF and PCM is shown in Fig. 8(b). The reaction and thermophysical properties of the

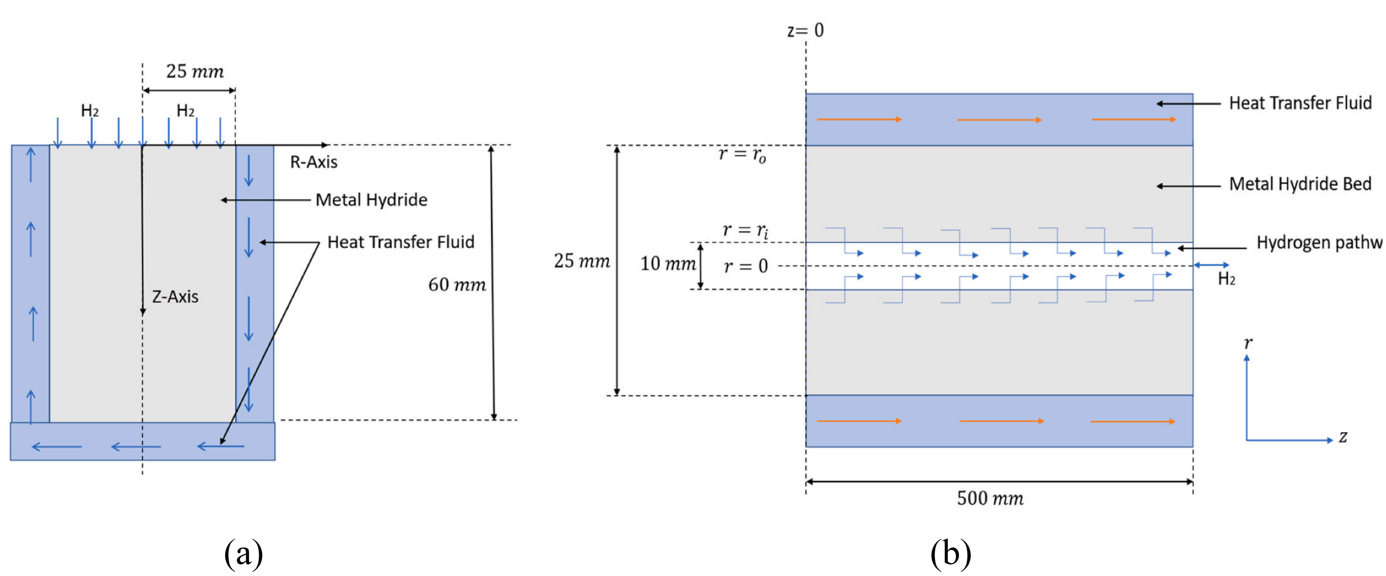

Fig. 7. 2D axisymmetric cross-sectional schematic of MH reactor with HTF (a) Considered by Jemni et al. [23,97] and others [5] (b) Considered by Yang et al. [60] and others [5].

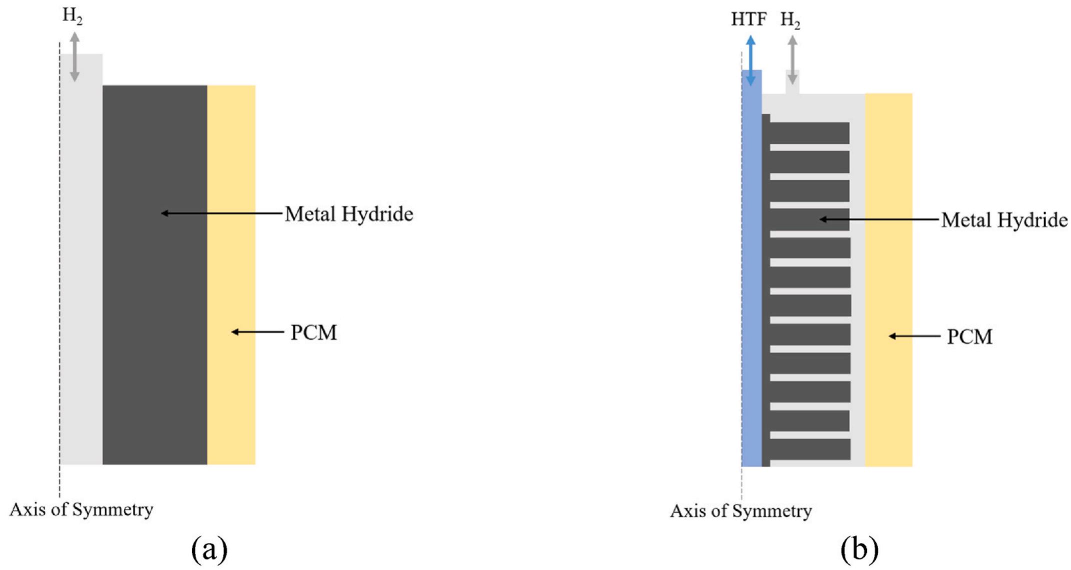

Fig. 8. Cross sectional schematic of MH reactor (a) Integrated with only PCM considered by Yao et al. [79] (b) Integrated with HTF and PCM considered by Ye et al. [50].

Table 6
Reaction and thermophysical properties of MH, PCM, HTF and H_2 for the Mg/MgH_2 + EG reactor modeled by Ye et al. [50,94].

Metal Hydride (Mg /MgH_2 + EG)	PCM ($NaNO_3$)	HTF (Dowtherm A)
$\rho_m = 1800$	$\rho_p = 2260 kg/m^3$	$\rho_h = 810\ kg/m^3$
$c_{p,m} = 1545\ J/\left(kg^{-1}\cdot K\right)$	$c_{p,p} = 1820\ J/\ kg\cdot K$	$c_{p,h} = 2350\ J/\ kg\cdot K$
$\lambda_m = 2\ W/(m\cdot K)$	$\lambda_p = 0.48\ W/(m\cdot K)$	$\lambda_h = 0.094\ W/(m\cdot K)$
$\varepsilon = 0.4$	$L = 174000\ J/kg$	$\mu_h = 2.15 \times 10^{-4}\ kg/mol$
$wt = 0.05$	$T_{sol} = 580.0\ K$	
$\Delta H = -75\ kJ/\ mol$	$T_{lid} = 581.0K$	
$\Delta S = -135.6J/\ (K\cdot mol)$		
$k_a = 10^{10}\ s^{-1}$	**Hydrogen Gas (H_2)**	
$k_d = 10\ s^{-1}$	$M = 0.002\ g/mol$	
$E_a = 130\ kJ/(\ K\cdot mol)$	$c_{p,g} = 14000\ J/kg/m^3$	
$E_d = 41\ kJ/(\ K\cdot mol)$	$\lambda_g = 0.24\ W/(m\cdot K)$	
$K = 5.75 \times 10^{-14}\ m^2$	$\mu_g = 8.9 \times 10^{-6}\ Pa\cdot s$	

bed, PCM, HTF, and the hydrogen gas considered in this work are shown in Table 6.

Utilizing the enthalpy-porosity formulation for the PCM, the heat transfer equation is expressed in terms of temperature and the total volumetric enthalpy (H) [50,141], where H, as a function of temperature T, is given by Eq. (72):

$$\frac{\partial H}{\partial t} = \nabla\cdot\left(\lambda_p \nabla(T)\right) \tag{71}$$

$$H(T) = \int_{T_{liq}}^{T} \rho_p c_{p,p} dT + \rho_p \cdot L_p \cdot f(T) \tag{72}$$

The solidification and dynamic melting behaviors can be characterized by the volumetric fraction of the liquid state PCM (f) that is given by Eq. (73).

$$f = \begin{cases} 0 & T < T_{sol} \\ (T - T_{sol})/(T_{liq} - T_{sol}) & T_{sol} \leqslant T \leqslant T_{liq} \\ 1 & T > T_{liq} \end{cases} \tag{73}$$

The results from this work [50] comparing the average temperature and average conversion rate of the MH bed and PCM before (MH-HTF

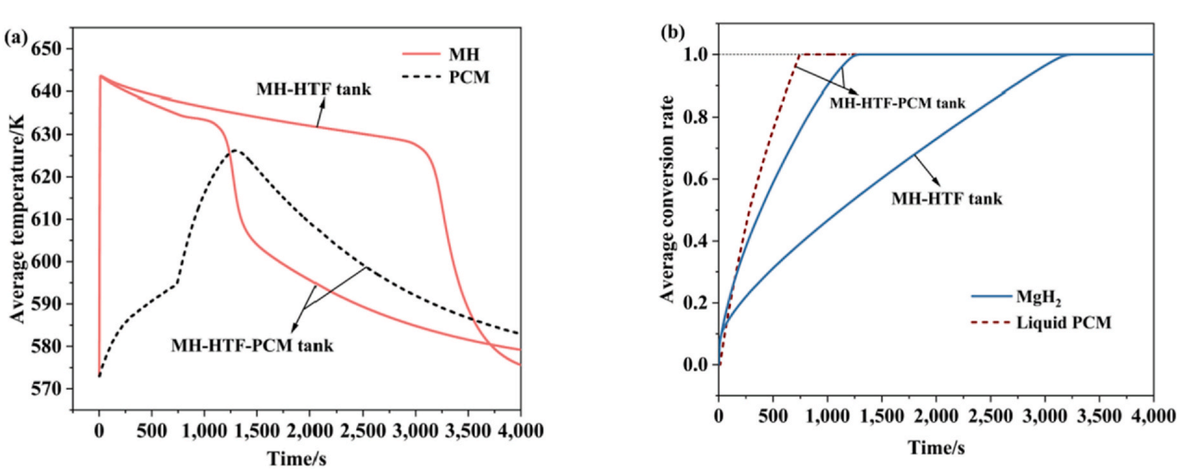

Fig. 9. (a) Comparison of the mean temperature for the MH bed and PCM during the hydrogen absorption, (b) Comparison of the mean conversion rate during the hydrogen absorption process (Reprinted with permission from Ref. [50]).

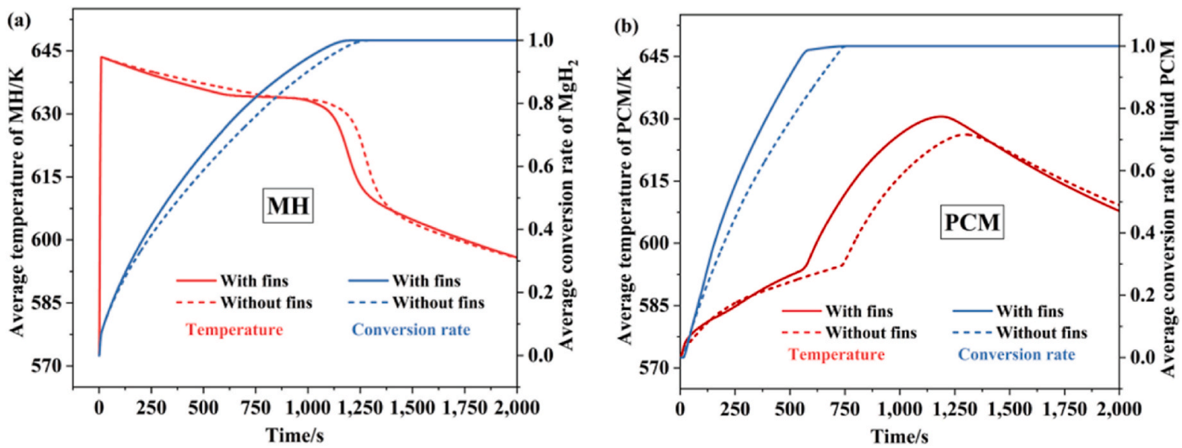

Fig. 10. (a) Comparison of the mean temperature and conversion rate of the MH bed between the tank with and without fins, (b) Comparison of the mean temperature and conversion rate of the PCM between the tank with and without fins (Reprinted with permission from Ref. [50]).

tank) and after incorporating PCM (MH-HTF-PCM tank) are shown in Fig. 9. Similarly, the influence of fins on these parameters is shown in Fig. 10.

4. Numerical methods and computaional tools

Different numerical methods can be utilized to solve the governing equations for MH systems, converting partial differential equations (PDEs) to a system of algebraic or ordinary differential equations (ODEs). The major numerical methods include the Finite Difference Method (FDM), which applies the Taylor series to discretize derivatives of dependent variables at various spatial and temporal points. The Finite Element Method (FEM) and Finite Volume Method (FVM) are also prominent. These methods use integral formulations to discretize the domain into sub-domains, accommodating complex geometries and discontinuous sources. FEM is particularly advantageous for handling Neumann boundary conditions without modifications to the equations [35], while FVM ensures good conservation properties and flexibility [30,91,151]. The Control Volume - Finite Difference Method (CV-FDM) is another notable approach, which discretizes space into subregions, solving equations based on neighboring points rather than control volumes [7].

Muthukumar and Ramana [151] developed a two-dimensional hydrogen desorption model using the finite volume method to predict hydrogen storage performance in a metal hydride bed filled with $MmNi_{6.4}Al_{0.4}$. Their study accounted for variations in the temperature of the cooling fluid, as well as the effects of hysteresis and plateau slope on the pressure-composition-temperature (PCT) characteristics of the MH alloy. Hasnain et al. [5,61,62,80] utilized a finite volume-based tool (known in Simulkade [152]) in MATLAB to develop an axially symmetric 2D model and conducted a comprehensive parametric analysis on different system parameters. Along with MATLAB, the finite-volume based tool 'Simulkade' is also available in Python, and Julia programming languages [152]. The Lattice Boltzmann method is also, sometimes, utilized in the literature for modeling MH hydrogen storage reactors. Valizadeh et al. [153] proposed a 2D mathematical model using the Lattice Boltzmann method to simulate transient heat and mass transfer during hydrogen desorption.

Various computational tools are extensively utilized for modeling and optimizing metal hydride (MH) hydrogen storage systems. The commercial software ANSYS Fluent, for example, is widely used for heat and mass transfer simulations in MH beds. Mellouli et al. [154] developed a computational model using ANSYS Fluent to analyze heat and mass transfer in a hydride bed integrated with a PCM for heat management. Their model compared cylindrical and spherical tank designs, demonstrating the higher efficiency of spherical tanks. Similarly, Chibani et al. [128] utilized ANSYS Fluent 15.0 to conduct detailed numerical simulations aimed at optimizing thermal management strategies. The simulations, performed with a time step of 0.10 s and a mesh comprising 8952 cells, were computationally very expensive requiring 22–25 days of computational time to achieve the numerical solution. Chung et al. [155] utilized ANSYS Fluent to create a 2D model predicting hydrogen storage performance and coolant behavior, testing tank designs with and without heat pipes and internal fins.

Alternatively, the open source software OpenFOAM is another prominent tool in this domain. Meneceur et al. [156] used OpenFOAM® CFD code coupled with finite volume method to develop a two-dimensional heat and mass transfer model for hydrogen desorption process in an annulus-disc reactor packed with Mischmetal (a commonly used intermetallic compound). They analyzed the impact of heating fluid temperature and outlet pressure on the thermal and reaction kinetics and achieved excellent agreement with experimental data through various numerical simulations.

Modern CFD software like FEMLAB, and COMSOL Multiphysics are instrumental in solving complex equations. COMSOL is noted for handling multiple physics modes simultaneously, making it highly suitable for combined mass, heat, and momentum transport in MH systems. An example includes a study where COMSOL Multiphysics 5.2 was used to develop a three-dimensional model of a Metal Hydride Reactor-Phase Change Material (MHR-PCM) system, incorporating modules for heat transfer in porous media and customized reaction dynamics [79]. COMSOL Multiphysics is specifically efficient for coupling thermal, fluid, and structural dynamics in MH storage systems. Mohammadshahi et al. [44,45] created a 3D model in COMSOL Multiphysics for a full absorption-desorption cycle and examined the impact of various parameters on metal hydride (MH) tank performance. Similarly, Aswin N. et al. [157] utilized COMSOL to model the thermal behavior of MH systems, while Tiwari et al. [158] recently developed a model and employed it for optimizing design parameters using response surface methods and sensitivity analysis.

MATLAB/Simulink is another tool in this context. Abdin et al. [57] developed a one-dimensional dynamic model of a metal hydride hydrogen storage tank using MATLAB/Simulink, analyzing the transient behavior during charging and discharging cycles. Additionally, GAMS (General Algebraic Modeling System) is utilized for optimization problems in MH systems. Talagañis et al. [159] applied GAMS to optimize design parameters, enhancing performance and efficiency of the storage systems.

5. Advancements in mathematical modeling: future research directions

There are several key avenues for future research pointing towards the advancement of mathematical modeling in metal hydride-based hydrogen storage systems to enhance the efficiency and reliability of these systems. Despite having rigorous attempts for modeling the complex behavior of MH based hydrogen storage system, current models still have simplifying assumptions to manage the computational complexities and numerical instabilities [7,51]. One of the major directions is the refinement of modeling techniques to capture the dynamic and transient behavior of metal hydride systems. There is a pressing need to develop more sophisticated models that account for factors such as pressure gradients, hydrogen velocity and diffusion, and variations in the thermophysical parameters during absorption and desorption processes. Further studies in this area will lead to achieving a balance between model complexity and accuracy that is needed for robust predictions.

In parallel, the exploration of multi-dimensional modeling represents a critical frontier. While one-dimensional and two-dimensional models are commonly employed, transitioning towards comprehensive three-dimensional models can provide more realistic results and open new areas for advancing metal hydride reactors [24,28,43,75]. This transition, however, demands significant computational resources, necessitating innovative approaches to address computational challenges without compromising accuracy [7]. Moreover, the consideration of local thermal equilibrium (LTE) and local thermal non-equilibrium (LTNE) models introduces another avenue for research [23,29,97]. Investigating the impact of assumptions related to temperature uniformity within the solid and gas phases can refine our understanding of the heat transfer mechanisms in metal hydride beds [103,104]. Future research should focus on validating and optimizing these assumptions to enhance the reliability of predictions. Similarly, modeling the physics of the phase change rather than making simple boundary conditions as well as applying the full Navier Stokes equations for the heat exchange fluids would allow for greater design flexibility.

Thermal management strategies form another critical aspect of future research. Efficient control of temperatures during absorption and desorption processes is essential for energy efficiency of the systems [52,160]. Exploring novel cooling techniques and the integration of phase change materials can significantly contribute to these efforts. In parallel with material considerations, enhancing heat and mass transfer in metal hydride systems is crucial. Innovative reactor designs, incorporating advanced heat exchangers and geometries, are important to improve overall system efficiency [1,31,51]. This involves optimizing designs to facilitate efficient energy exchange during the absorption and desorption processes.

Furthermore, the integration of advanced heat transfer models into mathematical frameworks represents an unexplored area. Exploring heat transfer mechanisms in conjunction with reaction kinetics can offer a more comprehensive understanding of the thermal management strategies required for efficient metal hydride-based hydrogen storage. This requires not only accounting for the effects of pressure drop and advective heat transfer but also investigating the impact of reactor designs and geometric parameters. The often-overlooked impact of hydrogen diffusion in metal hydride systems can also be addressed using Knudsen diffusion equations.

The role of data-driven and machine learning approaches cannot be ignored. Incorporating techniques, such as Physics-Informed Neural Networks (PINNs), into mathematical frameworks can enable the prediction of complex behaviors and identify correlations that traditional numerical models may struggle to capture [161,162]. Developing hybrid models that combine the strengths of traditional mathematical approaches with the flexibility of machine learning holds promise for more accurate and on demand predictions [163,164].

While the present review concentrates on the mathematical modeling of heat and mass transfer in metal hydride systems, readers seeking a broader treatment of unresolved challenges, emerging materials, and future perspectives in hydrogen storage are referred to recent comprehensive reviews. These include studies by Nivedhitha et al. [165] on carbon-based and metal hydride materials, Nivedhitha et al. [166] on mechanisms and challenges, Dewangan et al. [167] on outstanding issues and future prospects, Panigrahi et al. [168] on benefits and perspectives of storage methods and materials, Ward et al. [169] on technical challenges and future directions for hydride-based thermal storage, and Nemukula et al. [170] on sustainable hydrogen storage using metal hydrides.

Future research in the mathematical modeling of MH hydrogen storage systems should emphasize the development of advanced, multi-dimensional models that balance complexity and accuracy. Validating assumptions related to thermal equilibrium, integrating advanced heat transfer models, and exploring the potential of data-driven approaches represent crucial steps toward enhancing the predictive capabilities of these models. Embracing these research directions will not only deepen our understanding of metal hydride systems but also contribute to the optimization and design of efficient hydrogen storage technologies.

6. Conclusion

This review presents an in-depth and comprehensive analysis of mathematical models for MH-based hydrogen storage systems, focusing on heat and mass transfer processes. Growing demand for efficient and safer hydrogen storage has led to increased interest in metal hydrides as a viable alternative to traditional hydrogen storage approaches. With their high storage capacity and enhanced safety, metal hydrides stand out as a promising solution for hydrogen storage applications.

Starting from microscopic scale, this work explains the transition into macroscopic conservation equations in the porous MH bed based on effective medium theory and systematically explores different mathematical modeling approaches. It covers both local thermal equilibrium and non-equilibrium conditions, providing insights into the complexities of the MH-based hydrogen storage method. The paper provides a detailed analysis of modeling heat, mass, and momentum transport, considering factors such as reaction kinetics, the effects of considering various parameters, and different thermal management strategies. The discussion on assumptions, simplifications, and variables considered in various models, equilibrium pressure, and enhanced heat transfer through HTF and PCM contributes to a holistic understanding of the challenges and opportunities in designing efficient metal hydride systems offering a valuable resource for experts and emerging researchers in the field.

Looking ahead, the paper outlines future research directions to advance hydrogen storage technologies. It underscores the significance of further investigations into thermal management strategies, reactor configurations, and the development of accurate models for predicting the behavior of MH reactors under varying operating conditions. This review serves as a roadmap for future studies, guiding researchers towards enhancing the efficiency, safety, and practicality of MH-based hydrogen storage. Ultimately, the integration of rigorous mathematical modeling with experimental validation will contribute to the continued progress and widespread adoption of metal hydrides in the quest for sustainable and eco-friendly hydrogen storage solutions.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT to enhance the language and readability. Following the use of this tool, the content was carefully reviewed and edited as necessary. The authors assume full responsibility for the content of this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rser.2025.116294.

Data availability

No data was used for the research described in the article.

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